

REFRESHER READING

2024 CFA® PROGRAM • LEVEL 3

Portfolio Management

Risk Management for Individuals

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LEARNING OUTCOMES

Mastery	The candidate should be able to:
☐	compare the characteristics of human capital and financial capital as components of an individual's total wealth
☐	discuss the relationships among human capital, financial capital, and economic net worth
☐	describe an economic (holistic) balance sheet
☐	discuss risks (earnings, premature death, longevity, property, liability, and health risks) in relation to human and financial capital
☐	describe types of insurance relevant to personal financial planning
☐	describe the basic elements of a life insurance policy and how insurers price a life insurance policy
☐	discuss the use of annuities in personal financial planning
☐	discuss the relative advantages and disadvantages of fixed and variable annuities
☐	discuss how asset allocation policy may be influenced by the risk characteristics of human capital
☐	recommend and justify appropriate strategies for asset allocation and risk reduction when given an investor profile of key inputs

INTRODUCTION

1

Risk management for individuals is a key element of life-cycle finance, which recognizes that as investors age, the fundamental nature of their total wealth evolves, as do the risks that they face. **Life-cycle finance** is concerned with helping investors achieve their goals, including an adequate retirement income, by taking a holistic view of the

individual's financial situation as he or she moves through life. Individuals are exposed to a range of risks over their lives: They may become disabled, suffer a prolonged illness, die prematurely, or outlive their resources. In addition, from an investment perspective, the assets of individuals could decline in value or provide an inadequate return in relation to financial needs and aspirations. All of these risks have two things in common: They are typically random, and they can result in financial hardship without an appropriate risk management strategy. Risk management for individuals is distinct from risk management for corporations given the distinctive characteristics of households, which include the finite and unknown lifespan of individuals, the frequent preference for stable spending among individuals, and the desire to pass on wealth to heirs (i.e., through bequests). To protect against unexpected financial hardships, risks must be identified, market and non-market solutions considered, and a plan developed and implemented. A well-constructed plan for risk management will involve the selection of financial products and investment strategies that fit an individual's financial goals and mitigate the risk of shortfalls.

In this reading, we provide an overview of the potential risks to an individual or household, an analysis of products and strategies that can protect against some of these risks, and a discussion regarding the selection of an appropriate product or strategy. Following the introduction, Section 2 provides an overview of human and financial capital. Sections 3–5 address the process of risk management, the financial stages of life for an individual, the economic (or holistic) balance sheet, and individual risks and risk exposures. Sections 6–10 discuss the types of products relevant to financial planning, including insurance and annuities. Sections 11–12 contain an insurance program case study and insights on implementing risk management solutions for individuals.

2

HUMAN CAPITAL, FINANCIAL CAPITAL, AND ECONOMIC NET WORTH



compare the characteristics of human capital and financial capital as components of an individual's total wealth



discuss the relationships among human capital, financial capital, and economic net worth

To better understand the financial health of an individual—and how to manage the risks faced by that individual—we can use an **economic balance sheet** (or **holistic balance sheet**). We discuss the economic balance sheet in more detail later in the reading, but it is important to note here that an individual's assets are made up of two primary components, **human capital** and **financial capital**, which present unique risk management challenges.

Ibbotson, Milevsky, Chen, and Zhu (2007) define human capital as the net present value of an investor's future expected labor income weighted by the probability of surviving to each future age. Financial capital includes the tangible and intangible assets (outside of human capital) owned by an individual or household. For example, a

home, a car, stocks and bonds, a vested¹ retirement portfolio, and money in the bank are all examples of an individual's financial capital (or financial assets). In this section, both human capital and financial capital are explored in greater detail.

Human Capital

Advances in human capital theory have revolutionized how economists view the household risk management process. Conceptually, future wages or earnings can be thought of as analogous (in a rough sense) to future interest or dividend payments that flow from an individual's work-related skills, knowledge, experience, and other productive attributes that can be converted into wage income—or human capital. Because human capital provides a significant stream of income over decades, its present value is a significant part of most working households' total wealth portfolio. In fact, human capital is often the dominant asset on a household's economic balance sheet. From a risk management perspective, it is critical to understand the approximate total monetary value of an individual's human capital, the investment characteristics of the individual's human capital (i.e., whether the capital is more stock-like or bond-like), and how the approximate value of an individual's human capital relates to the value of the individual's financial capital. Here, we focus on estimating the approximate monetary value of an individual's human capital.

Given that future earnings for many workers are relatively stable over time, earnings can often be compared with the income one might receive from a bond. This analogy is useful because, similar to the way a financial analyst estimates the present value of a bond by discounting future cash flows, we can estimate human capital by discounting the expected future cash flows generated from wages or other income sources. Conceptually, individuals rent out or lease their human capital in the marketplace in exchange for an ongoing income that is a function of the state of the labor market. Some professions will receive a higher rental value (wage rate or salary) than others. Similarly, some professions will see their rental value fluctuate more with changes in the labor market environment.

Estimating the value of human capital is a complex process because the true value cannot be known. One simple approach is to use a discount rate that reflects the risk associated with the future cash flows (i.e., wages). Government employment and teaching are examples of professions that generally lead to relatively stable growth of future cash flows; in these cases, the human capital value would be estimated using a lower discount rate (to reflect the higher degree of certainty). Conversely, investment banking and racecar driving are examples of professions that may experience unstable and less secure future cash flows, so the value of human capital of investment bankers and racecar drivers would be based on a higher discount rate (to reflect the additional risk associated with their professions).

Equation 1 can be used to estimate the value of an individual's human capital today, at Time 0 (HC_0), where w_t is the income from employment in year t , r is the appropriate discount rate, and N is the length of working life in years. Working life typically ends at retirement, although it could also be based on the number of years an individual can potentially work (e.g., a 70-year-old may be retired but still have some remaining human capital that could be traded for income in the labor market):

$$HC_0 = \sum_{t=1}^N \frac{w_t}{(1+r)^t} \quad (1)$$

¹ Vesting refers to ownership of retirement or pension benefits. Once benefits are vested, they belong to the beneficiary. Benefits that are not fully vested can be forfeited or reduced if the individual does not meet future conditions (e.g., if the individual terminates employment before the required number of years of service to the organization).

This simple model in Equation 1 can be expanded using Equation 2, where we define the wage in time period t as a product of the wage in period $t - 1$ and the sum $(1 + g_t)$. That is, the wage in a given period is equal to the previous year's wage increased by g percent (the annual wage growth rate, in nominal terms). We can also modify the discount rate to be the sum of the nominal risk-free rate r_f and a risk adjustment y based on occupational income volatility. Similar to our example earlier contrasting the overall stability of labor income for government workers and teachers to that of investment bankers and racecar drivers, this adjustment recognizes the fact that the income from different professions can vary significantly. The risk adjustment should consider the inherent stability of the income stream as well as the possibility that the income stream will be interrupted by job loss, disability, or death that may be completely unrelated to the type of employment. Additionally, we incorporate mortality, where $p(s_t)$ is the probability of surviving to a given year (or age). Equation 1 uses a simplifying assumption that $p(s_t) = 1$ for each year until retirement (i.e., the individual will survive to retirement with certainty). Using these additional factors, Equation 2 can be restated as:

$$HC_0 = \sum_{t=1}^N \frac{p(s_t) w_{t-1} (1 + g_t)}{(1 + r_f + y)^t} \quad (2)$$

EXAMPLE 1

Estimating Human Capital

1. Identify the key assumptions required to estimate an individual's human capital.

Solution:

Human capital can be calculated by using the following formula:

$$HC_0 = \sum_{t=1}^N \frac{p(s_t) w_{t-1} (1 + g_t)}{(1 + r_f + y)^t}$$

where

$p(s_t)$ = the probability of surviving to year (or age) t

w_t = the income from employment in period t

g_t = the annual wage growth rate

r_f = the nominal risk-free rate

y = risk premium associated with occupational income volatility

N = the length of working life in years

ESTIMATING THE PRESENT VALUE OF HUMAN CAPITAL

Using Equation 2, we briefly demonstrate how to estimate the present value of an individual's human capital. John Adam is 60 years old and plans on retiring in 5 years. Adam's annual wage is currently \$50,000 and is expected to grow 2% per year. The risk-free rate is 4%. Adam works in a job with a moderate degree of occupational risk; therefore, we assume a risk adjustment based on occupational income volatility of 3%. There is a 99% probability that Adam survives the first year, a 98% probability that he

survives the second year, and probabilities of 98%, 97%, and 96% for the following years, respectively. Given this information and using Equation 2, what is the present value of Adam's human capital?

Risk-free rate	= 4%
Income volatility adjustment	= 3%
Total discount rate	= 7%

Year	Wages (2% annual growth)	Present Value of Wages ^a	Probability of Survival	Probability Weighted Wages ^b
1	\$51,000	\$47,664	99%	\$47,187
2	\$52,020	\$45,436	98%	\$44,527
3	\$53,060	\$43,313	98%	\$42,447
4	\$54,122	\$41,289	97%	\$40,050
5	\$55,204	\$39,360	96%	\$37,786
Total value of human capital				\$211,997

^a This column illustrates "Wages" discounted by 7% as indicated by the discount rate shown. For example: $\$47,664 = \$51,000/1.07$; $\$45,436 = \$52,020/1.07^2$; and so on.

^b The calculation for this column is as follows: $\$47,187 = \$47,664 \times 99\%$. A similar calculation is used for the following years.

How would the estimated value of Adam's human capital change if the wage growth rate were changed to 0%, the risk-free rate decreased to 2%, and the risk adjustment for occupational income volatility also decreased to 2% (using the same base wage and mortality estimates)?

Risk-free rate	= 2%
Income volatility adjustment	= 2%
Total discount rate	= 4%

Year	Wages (No Growth)	Present Value of Wages	Probability of Survival	Probability Weighted Wages
1	\$50,000	\$48,077	99%	\$47,596
2	\$50,000	\$46,228	98%	\$45,303
3	\$50,000	\$44,450	98%	\$43,561
4	\$50,000	\$42,740	97%	\$41,458
5	\$50,000	\$41,096	96%	\$39,453
Total value of human capital				\$217,371

Reality is typically more complicated than models. Growth rates, nominal risk-free rates, risk adjustments, and mortality are not easily estimated. Additionally, wages do not tend to increase at a constant rate over an individual's lifetime, mortality and disability risk can reduce the value of human capital, and the average growth rate within occupations or even within the overall economy is unknown. In other words,

the future payout on human capital, like the future payout on many financial assets, is uncertain. The potential loss of human capital, particularly early in the life cycle, represents an important risk that must be considered. Life and disability insurance, which we discuss later in the reading, are examples of financial instruments that can be used to protect against a random loss in household earnings. As human capital diminishes later in the life cycle, other risks that threaten financial capital and increase spending needs rise in importance. Accordingly, strategies that reduce investment risk and protect against long-term health care expenses and long-life spending needs increase in importance.

Viewing human capital as an asset with its own risk and return characteristics allows us to develop a holistic investment strategy that includes tangible and intangible assets. A total wealth perspective combines human capital with financial capital and incorporates the concept of life-cycle planning (also discussed later) to develop a strategy that maximizes household welfare.

Financial Capital

Financial capital can be subdivided into various components besides tangible and intangible, such as personal assets and investment assets. Investment assets can be further differentiated into many subtypes with distinctive marketability, tax, and standalone risk characteristics. The relationships between the value of the various components of an individual's financial capital and the value of his or her human capital are important in investment and risk management decision making.

The approach used in financial accounting provides an excellent template for classifying the different financial assets owned by an investor. In financial accounting, the balance sheet includes a summary of all the assets owned by an entity, whether an individual or organization, at a given point in time. Assets are defined broadly as either current or non-current. Current assets are expected to be consumed over the following year; money in a checking account, for example, would be considered a current asset. For an individual, non-current assets—that is, all assets not classified as current assets—include such items as automobiles, real estate, and investments (such as stocks and bonds). Non-current assets differ for a company because they include such items as property, plant, and equipment, as well as intangible assets, such as goodwill.

The financial accounting approach to segmenting assets has important implications when assessing an individual's financial capital because different assets have different roles and each may be exposed to various types of risk. Broadly speaking, an individual's assets can be described as “personal” assets or “investment” assets; personal assets are consumed whereas investment assets are held for their potential to increase in value and fund future consumption. Some assets, such as real estate, can act as both a personal asset (shelter, as an alternative to renting) and an investment asset (to help fund retirement) for an individual.

EXAMPLE 2

Comparing Financial and Human Capital

1. Describe human capital and financial capital.

Solution:

Human capital is commonly defined as the mortality-weighted net present value of an individual's future expected labor income. Financial capital includes the tangible and intangible assets (outside of human capital) owned

by an individual or household. For example, a home, a car, stocks, bonds, a vested retirement portfolio, and money in the bank are all examples of an individual's financial capital (or financial assets).

Personal Assets

Personal assets are assets an individual consumes (or uses) in some form in the course of his or her life. Such assets may include automobiles, clothes, furniture, and even a personal residence. In many cases, personal assets are not expected to appreciate in value, and they are often worth more to the individual than their current fair market value.

As mentioned earlier, some assets, like real estate, could be considered a “mixed” asset with both personal and investment characteristics. Another potential example of a mixed asset is collectibles (such as jewelry, wine, stamps, and artwork), which will be discussed separately in a later section. Mixed assets can be especially desirable because they enable individuals to derive satisfaction (i.e., utility) from their current value as well as having the potential to increase in value over time.

CLASSIFYING PRIVATE ACCRUED DEFINED BENEFITS AND GOVERNMENT RETIREMENT BENEFITS

When separating an individual's total wealth into human capital and financial capital, accrued defined benefits from private pension and government retirement plans—such as the Canada Pension Plan and Old Age Security Pension, the Age Pension in Australia, the mandatory state pension in Germany, and Social Security in the United States—can potentially be classified as either human capital or financial capital. Some practitioners note that accrued defined benefits and government pension benefits are typically a form of deferred labor income, and thus, they prefer to classify these benefits as human capital. Others find it more intuitive to think of accrued defined benefits and social security as a form of human capital that has been converted into a financial asset. In this reading, we classify accrued defined benefits and government pension benefits as components of financial capital.

Investment Assets

Investment assets are the components of an individual's wealth that are often the easiest to identify and typically receive the majority of the attention from financial planners and investment professionals. Investment assets extend beyond relatively tangible investment assets (such as a liquid portfolio) to include less tangible assets (such as an accrued defined benefit pension).

One criterion for subdividing investment assets is marketability, which describes how easy it is to trade an asset. We subdivide marketable assets into publicly traded and non-publicly traded segments, and we define non-marketable assets as those without any ready market (e.g., human capital).

Traditional portfolio construction generally focuses entirely on publicly traded marketable assets, like stocks and bonds, with optimization determining the weights allocated to marketable assets. This approach often ignores the existence of marketable assets that are not publicly traded as well as other non-marketable assets “owned” by the individual (e.g., human capital). In reality, each asset has important risk characteristics that should be considered. In the absence of a generalized framework that can estimate the risk and return of all of the components of an individual's total wealth and their correlations, one must understand the inherent risk and return characteristics of the non-marketable assets and make informed judgments when constructing a holistic portfolio. We will explore this concept more fully later in the reading.

Publicly Traded Marketable Assets

Traditional balance sheets tend to emphasize publicly traded marketable assets because their value and risk characteristics are generally easier to estimate than those of non-publicly traded assets. Publicly traded marketable assets include money market instruments, bonds, and common and preferred equity.

Non-Publicly Traded Marketable Assets

Non-publicly traded marketable assets include real estate, some types of annuities, cash-value life insurance, business assets, and collectibles.

Real Estate

Real estate—or direct real estate, as it is sometimes called to distinguish it from real estate investment trusts (REITs)—is typically among the largest assets owned by an individual. In many countries, home ownership is common, although the level of home ownership varies materially by country. For example, in Germany, approximately half of households own a home, whereas in China, the number is closer to 90%. To purchase a home, many individuals obtain a mortgage loan. It is common for the home buyer to contribute some percentage of the home's value (e.g., 20%) as a down payment to mitigate some of the risk to the lender. The term of the mortgage loan can vary (e.g., 15 years, 30 years), as can the mortgage's interest rate (which can be either fixed or floating). Mortgage payments are often the largest fixed obligation of homeowners, especially during the early years of a mortgage loan. Mortgages present a unique risk for homeowners because they create a leveraged exposure in a home. For example, a 20% down payment (80% mortgage loan) implies that for any given change in the value of the home, the change in the equity (value less the mortgage loan) of the home will be five times greater than the change in the value of the home. Mortgage loans are either recourse or non-recourse, and the status varies by region. With recourse mortgages, if the borrower defaults on the mortgage, the lender has the right to recover from the borrower any amount due on the loan, whereas non-recourse loans prevent the lender from recovering any further amount from the borrower. Non-recourse loans are thus riskier for lenders because the only available collateral for the loan is the home. As a result, non-recourse loans generally have higher interest rates and/or higher borrower credit standards than recourse loans.

Annuities

Annuities are effectively a private defined benefit pension for which an insurance company has guaranteed, or will guarantee, income for life or over some fixed period for the beneficiary (called the annuitant). The estimated balance sheet value of an annuity is comparable to that of a defined benefit pension with a discount for potential insolvency risk, which is difficult to eliminate through diversification or a market hedge. Annuities will be discussed in more detail later in the reading.

Cash-Value Life Insurance

A variety of types of life insurance are available, including cash-value life insurance, for which the policy not only provides protection upon a death but also contains some type of cash reserve. This form of insurance usually combines life insurance protection with some type of cash accumulation vehicle. Some insurance policies allow the purchaser to invest in relatively aggressive investments, such as equities, although more conservative investments, such as bonds, are generally more common. Life insurance products will also be discussed at greater length later in the reading.

Business Assets

Business assets can represent a significant portion of the total wealth of an individual, especially a self-employed individual. A variety of unique considerations are involved in investing for business owners because their total capital may be very closely tied

to the overall performance of the business (i.e., if the business does poorly, it affects not only the value of the business, but also the owner's earnings as well). The value of business assets may best be estimated through recent sales of comparable private businesses within the same industry—often as a multiple of net income or net income with various adjustments (e.g., EBITDA). The value of business assets may vary based on market conditions and will often correlate with other financial assets within a household portfolio. This potential correlation is an important consideration in the risk management process for individuals, particularly small business owners.

Collectibles

Collectibles include such items as stamps, paintings, wine, and precious metals (e.g., coins). The value of these assets is often set by auction markets or specialized dealers and involves substantial transaction costs. Collectibles may also provide a flow of utility for the owner. For example, in addition to benefiting from the potential price increase of a painting, the owner is able to display the painting in his or her home and view it daily.

Non-Marketable Assets

The most significant non-marketable financial assets are pensions, whether from a private employer or from a governmental organization. In this section, we consider both types of pensions.

Employer Pension Plans (Vested)

There are a variety of retirement plan types across the globe. These accounts can generally be described as either employee-directed savings plans, in which contribution amounts and investments are controlled by the individual (and not guaranteed), or traditional pension plans, which guarantee some level of retirement benefits, typically based on past wages. We include only vested pension benefits as financial assets, because unvested pension benefits are typically contingent on future work and are thus considered to be part of human capital.

The value of a vested traditional defined benefit pension from an employer can be estimated by determining the mortality-weighted net present value of future benefits. The mortality-weighted net present value at Time 0 (now), $mNPV_0$, can be estimated using Equation 3, which is reasonably equivalent to Equation 1. Equation 3 is based on the future expected vested benefit (b_t), the probability of surviving until year t [$p(s_t)$], and a discount rate (r). The discount rate should vary based on the relative riskiness of the future expected benefit payment—that is, the rate will be higher for riskier future benefit payments—and should reflect whether the benefit is in nominal or real terms:

$$mNPV_0 = \sum_{t=1}^N \frac{p(s_t) b_t}{(1+r)^t} \quad (3)$$

Estimating an appropriate discount rate to use in valuing a pension can be quite complex, although it is generally less complex than estimating the discount rate to use in valuing an individual's human capital. There are a number of factors to consider in determining the pension discount rate. As a starting point, one should consider the health of the plan (e.g., its funding status, where the value of the plan's liability is estimated using an appropriate market-based discount rate), the credit quality of the sponsoring company, and any additional credit support. If the company in question has long-term bonds, the yield on the bonds can provide a proxy for an appropriate discount rate. As one example of credit support, the Pension Protection Fund (PPF) was established in the United Kingdom as part of the Pensions Act 2004 to guarantee continued payment of most UK defined benefit pension plans should the employer become insolvent. The existence of PPF insurance helps to decrease the payout risk (and accompanying discount rate) for eligible UK pension plans.

Government Pensions

Government pensions are similar to employer pension plans but are generally more secure (in those countries with a high degree of creditworthiness). As with employer pension plans, the vested or accrued benefit amount can be estimated by calculating the mortality-weighted net present value. Given the guaranteed nature of these benefits, government pensions can be considered relatively bond-like. For example, in the United States, retiree government pension benefits (called Social Security retirement benefits) can be thought of as a government bond with benefits indexed to inflation (because Social Security retirement benefits usually increase annually based on inflation). This inflation adjustment is consistent with securities called Treasury Inflation-Protected Securities (TIPS).

Regardless of the domicile, one should consider the financial health of the government entity sponsoring the defined benefit plan as well as the legal framework and any accompanying political risk at the country level.

Account Type

Financial capital is often held in account types that have different tax attributes. Although these account types (and the potential tax benefits surrounding them) vary materially by country, the accounts can generally be described as taxable, tax-deferred, or non-taxable. A taxable account is one for which taxes are due annually on the realized gains, dividends, and/or interest income. A tax-deferred account is one for which taxes on any gains are deferred until some future date, such as when a withdrawal is made from the account. A non-taxable account is one for which taxes are never due, no matter how much the account grows.

Economic Net Worth

An individual's *networth* consists of the difference between traditional assets and liabilities that are reasonably simple to measure, such as investment assets, real estate, and mortgages. **Economic net worth**, however, extends net worth to include claims to future assets that can be used for consumption, such as human capital and the present value of pension benefits. When we refer to economic net worth in this reading, we refer to the more holistic accounting of resources that can be used to fund future consumption for the purpose of financial planning over the life cycle.

3

A FRAMEWORK FOR INDIVIDUAL RISK MANAGEMENT

This section contains an overview of the important considerations when developing an effective risk management plan for an individual. First, a risk management strategy for individuals is introduced. Next, the primary financial stages of the life of an individual are discussed. We then incorporate the human capital and financial capital concepts developed in Section 2 into an individual's economic (or holistic) balance sheet, explaining how key components of that balance sheet develop over time. Finally, we identify some of the primary risks to an individual and how they evolve during an individual's lifetime.

The Risk Management Strategy for Individuals

In general, *risk management* for individuals is the process of identifying threats to the value of household assets and developing an appropriate strategy for dealing with these risks. The risk management strategy provides a framework that allows a household to decide when to avoid, reduce, transfer, or self-insure those risks. There are typically four key steps in the risk management process:

1. Specify the objective.
2. Identify risks.
3. Evaluate risks and select appropriate methods to manage the risks.
4. Monitor outcomes and risk exposures and make appropriate adjustments in methods.

Specify the Objective

The overarching objective of individual risk management is to maximize household welfare through an appropriate balance of risk and safety. Risk represents a possible decrease in future spending caused by unexpected events, such as a market crash, a physical disability, the premature death of a primary earner, or health care expenses. As with investments, this objective is achieved by deciding how much risk a household is willing to bear in order to achieve its long-run spending goals.

Identify Risks

Households face a significant number of risks, including earnings, premature death, longevity, property, liability, and health risks. These risks will be discussed at length in subsequent sections. Each of these risks is associated with a potential loss of financial and/or human capital, and individuals should address each of them to determine how best to address the possibility of loss.

Evaluate Risks and Select Appropriate Methods to Manage the Risks

The existence of a risk exposure does not necessarily require the purchase of an insurance product. The appropriate risk management strategy considers the magnitude of the risk and the range of options available to address that risk. Section 11 will explain the choice among the four techniques of risk avoidance, risk reduction, risk transfer, and risk retention. *Risk avoidance* involves avoiding a risk altogether. For example, one way to avoid the risk to human and financial capital from riding a motorcycle is to simply not own or ride one. *Risk reduction* involves mitigating a risk by reducing its impact on an individual's welfare, either by lowering the likelihood that it will occur or by decreasing the magnitude of loss (for example, by wearing a helmet when riding a motorcycle). *Risk transfer* involves transferring the risk: The use of insurance and annuities to transfer risk to insurers will be discussed later in the reading. *Risk retention* involves retaining a risk and thus maintaining the ability to finance the cost of losses; when funds are set aside to meet potential losses, the individual is said to *self-insure*.

Monitor Outcomes and Risk Exposures and Make Appropriate Adjustments in Methods

Once the appropriate risk management method has been selected, risks must be monitored and updated as the household moves through its life cycle. It is advisable to annually review an insurance/risk management program, including all the ongoing risk exposures and risk management methods. As an individual's goals and personal and financial situation change, these changes will affect risk exposures and optimal risk management strategies. In addition to an annual review, every life change—such as a birth, marriage, inheritance, job change, relocation, divorce, or death—should trigger a review of the risk management plan.

Financial Stages of Life

Individuals tend to follow a predictable pattern during their lifetimes: They invest in education early in life, embark on a career, start families, accumulate assets, fund growing household expenses, transition into retirement, and ultimately pass on wealth through bequests. In each of these life-cycle stages, the household faces unique goals and risks that require appropriate investment and risk management strategies.

Defining financial stages of life in clear and concise terms does pose a challenge because all individuals are different; however, financial stages are a useful construct when thinking about risk management and the optimal forms of insurance and other products to consider at different ages. Therefore, we divide the financial stages of life for adults into the following seven periods:

- Education phase
- Early career
- Career development
- Peak accumulation
- Pre-retirement
- Early retirement
- Late retirement

Education Phase

The education phase occurs while an individual is investing in knowledge (or human capital) through either formal education or skill development. In theory, the education phase could begin as early as when an individual starts primary school, but this phase typically involves the period when the individual starts developing more specific human capital by attending college or trade school or undertaking an apprenticeship. In some cases, an individual in the education phase may be largely financially dependent on his or her parents or guardians and have little, if any, accumulated financial capital. There is generally little focus on savings or risk management at this point; however, some individuals in this phase may already have families and could benefit from products, such as life insurance, that hedge against the risk of losing human capital.

Early Career

The early career phase normally begins when an individual has completed his or her education and enters the workforce. This stage may begin as early as age 18 (16 in some countries) or as late as the late 20s (or even early 30s), depending on the level of education attained, and generally lasts into the mid-30s. During this period, the individual often marries, perhaps has young children, may purchase a home, and usually begins to save for their children's college expenses. Sometimes, a career-related relocation occurs that could have negative short-term financial implications. Significant family and housing expenses may not allow for much retirement savings. Insurance may be especially valuable during this phase because human capital represents such a large proportion of total wealth and family members are highly dependent on the human capital of one or two individuals to fund expected future consumption.

Career Development

The career development phase normally occurs during the 35–50 age range and is often a time of specific skill development within a given field, upward career mobility, and income growth. This phase often includes accumulation for the children's college educations as well as expenditures for college. Concern intensifies about retirement income planning and financial independence. Higher earners will begin building wealth

beyond education and retirement objectives and may make large purchases, such as a vacation home, or travel extensively. Retirement saving tends to increase at a more rapid pace during this phase compared with the early career phase.

Peak Accumulation

In the peak accumulation phase, generally during the ages of 51–60, most people either have reached or are moving toward maximum earnings and have the greatest opportunity for wealth accumulation. This phase may include accumulating funds for other goals and objectives, but it is usually a continuation of retirement income planning, coordination of employee benefits with investment and retirement strategies, and travel. Investors following a life-cycle portfolio strategy will begin to reduce investment risk to emphasize income production for retirement (particularly near the end of this period) and become increasingly concerned about minimizing taxes, given higher levels of wealth and income. There is also potentially more career risk in this phase because if an individual were to lose his or her job, it might be relatively difficult for that individual to find another job with similar pay.

Pre-retirement

The pre-retirement phase consists of the few years preceding the planned retirement age, and it typically represents an individual's maximum career income. Many people in this phase continue to restructure their portfolios to reduce risk and may consider investments that are less volatile. There is further emphasis on tax planning, including the ramifications of retirement plan distribution options.

Early Retirement

The early retirement phase in the cycle is generally defined as the first 10 years of retirement and, for successful investors, often represents a period of comfortable income and sufficient assets to meet expenses. For individuals who are forced to retire because of injury or unemployment, this time may be one of shifting expectations and may involve changing to a lifestyle more commensurate with the individual's savings. This is generally the most active period of retirement and is when an individual is less likely to suffer from cognitive or mobility limitations. The primary objective of the retiree is to use resources to produce activities that provide enjoyment. Some retirees seek a new career, and many will look for a job (part time or full time) that has less stress. It is important to note that upon entering retirement, the need for asset growth does not disappear. For many households, the length of retirement could exceed two decades; given this potential horizon, it is important to continue taking an appropriate level of investment risk in retirees' portfolios.

Late Retirement

The late retirement phase is especially unpredictable because the exact length of retirement is unknown. This uncertainty about longevity for a specific individual is known as longevity risk, which is the risk that an individual outlives his or her financial resources in retirement. Physical activity typically declines during this phase, as does mobility. Although many individuals live comfortably and are in good health until their final days, others experience a long series of physical problems that can deplete financial asset reserves. Cognitive decline can present a risk of financial mistakes, which may be hedged through the participation of a trusted financial adviser or through the use of annuities. Annuities will be discussed in more detail later in the reading.

Two additional concerns may be appropriate to any financial stage. First, depending on the family situation, the need to provide for long-term health care may become apparent. Second, some people may need to devote resources to care for parents or a disabled child for an extended period of time.

EXAMPLE 3**Financial Stages of Life**

1. From a personal financial planning standpoint, what are typical characteristics of someone in the “peak accumulation” phase?

Solution:

An individual in the peak accumulation phase of the life cycle would typically have the following characteristics:

- Approximate age of 51–60
- Maximum earnings and opportunity for wealth accumulation
- Increased interest in retirement income planning
- Greater emphasis on stability and less emphasis on growth in the investment portfolio
- Greater concern about tax strategies
- Increased concern about losing employment because it may be more difficult to find new employment

4**THE INDIVIDUAL BALANCE SHEET**

describe an economic (holistic) balance sheet

A traditional balance sheet includes assets and liabilities that are usually easy to quantify. Our purpose in developing an individual balance sheet is to more comprehensively represent the assets available to fund life-cycle consumption and for wealth preservation and transfer bequests. The primary value of a balance sheet in this context is to illustrate the magnitude of risk exposures for an individual. This perspective is particularly important for individuals who are in life-cycle stages during which human capital is a significant share of overall wealth and for individuals who hold claims on pension assets that grow in value later in the life cycle.

In this section, we attempt to provide a more complete picture of an investor's wealth through the use of an economic balance sheet (or holistic balance sheet), which we initially mentioned in Section 2. Such a balance sheet provides a useful overview of the individual's total wealth portfolio, supplementing traditional balance sheet assets with human capital and pension wealth and expanding liabilities to include consumption and bequest goals. These additional liabilities are important because they often represent leverage created in order to gain access to assets, such as the cost of education to create human capital. They also represent regular payment obligations that may influence the optimal amount of portfolio liquidity and investment risk. Human capital and pension wealth are important because they represent expected income flows that can be drawn on to fund future consumption.

Traditional Balance Sheet

The simplest balance sheet for an individual investor includes recognizable marketable assets and liabilities. Assets include any type of investment portfolio, retirement portfolio (or plan), real estate, and other tangible and intangible items of value. Liabilities include mortgage debt, credit card debt, auto loans, business debt, and student loans. An example of a simple balance sheet (or statement of net worth) is shown in Exhibit 1, where the assets are netted against the liabilities to determine the net worth of the individual.

Exhibit 1: Traditional Balance Sheet as of 31 December 2014

Assets		Liabilities	
Liquid Assets		Short-Term Liabilities	
Checking account	€35,000	Credit card debt	€25,000
Certificates of deposit	€100,000	Total short-term liabilities	€25,000
Total liquid assets	€135,000		
Investment Assets		Long-Term Liabilities	
Taxable account	€750,000	Car loan*	€25,000
Retirement plan	€600,000	Home mortgage	€500,000
Cash value of life insurance	€25,000	Home equity loan	€90,000
Total investment assets	€1,375,000	Total long-term liabilities	€615,000
Personal Property			
House	€2,200,000		
Cars	€160,000		
House contents	€150,000		
Total personal property	€2,510,000		
Total Assets	€4,020,000	Total Liabilities	€640,000
		Net Worth	€3,380,000

* Note: A portion of the car loan would likely be short term, but to simplify, we included the entire loan as a long-term liability.

The net value of an asset, or its equity, is calculated by subtracting liabilities associated with that asset from the gross value. For example, an individual may own a home worth £1 million, but if that individual has a £900,000 mortgage, the equity in the home would be only £100,000 (ignoring any additional intangible benefits associated with home ownership).

It should be noted that this traditional balance sheet includes those assets that can be valued easily but ignores other individual assets that are material, such as human capital and pension benefits. For individuals in the earlier life-cycle stages, human capital is larger than other assets on the balance sheet. For those who are eligible to receive a guaranteed retirement income stream, the present value of these assets is significant and can be of great value to older individuals. Although non-marketable

and difficult to value precisely, human capital and retirement benefits are extremely important when planning the optimal use of assets and the repayment of liabilities over a life cycle.

Economic (Holistic) Balance Sheet

The primary goal of an economic (holistic) balance sheet is to arrive at an accurate depiction of an individual's overall financial health by accounting for the present value of all available marketable and non-marketable assets as well as all liabilities. This view allows an individual to map out the optimal level of future consumption and non-consumption goals (such as bequests or other transfers) given the resources that exist today and those that are expected in the future. Although a traditional balance sheet provides information about marketable assets that exist today, it offers limited insight into how these assets should be used to maximize the expected lifetime satisfaction of the individual (a concept economists call “utility”). An economic balance sheet allows an individual to anticipate how available resources can be used to fund consumption over the remaining lifetime.

Exhibit 2 provides a simplified example of an economic balance sheet, which is an expanded version of the traditional balance sheet in Exhibit 1. The traditional assets and liabilities are condensed from the traditional balance sheet in Exhibit 1, with the present value of human capital and pensions added as assets and the present value of lifetime consumption and bequests added as liabilities. For further simplification purposes, we assume that all the assets and liabilities in Exhibit 1 are already calculated at their present value.

Exhibit 2: Economic (Holistic) Balance Sheet as of 31 December 2014

Assets		Liabilities	
Financial capital	€4,020,000	Debts	€640,000
Liquid assets		Credit card debt	
Investment assets		Car loan	
Personal property		Home mortgage	
		Home equity loan	
Human capital	€1,400,000	Lifetime consumption needs (present value)	€4,200,000
Pension value	€500,000		
		Bequests	€400,000
Total Assets	€5,920,000	Total Liabilities	€5,240,000
		Economic Net Worth	€680,000

An economic balance sheet that includes the present value of non-marketable assets (e.g., human capital and pensions) and liabilities (e.g., consumption needs and bequests) provides a much more accurate baseline from which to maximize the expected utility of future consumption. Assessing pension and human capital value can also be useful when setting consumption or bequest goals because these assessments provide a more accurate estimation of the future trade-offs an individual will make. Younger

households with greater human capital, in addition to spending more to protect the value of this human capital early in the life cycle, will be able to plan for more generous retirement savings goals than households with comparatively lower human capital.

The total economic wealth of an individual changes throughout his or her lifetime, as do the underlying assets that make up that wealth. The total economic wealth of younger individuals is typically dominated by the value of their human capital because younger individuals have not had as much time to save and accumulate financial wealth. As individuals grow older, they are likely to save some of their earnings and will accumulate financial capital. The total value of human capital and the total value of financial capital tend to be inversely related over time as individuals attempt to smooth consumption through borrowing, saving, and eventual spending. When human capital is depleted, an absence of financial capital would result in no wealth to fund an individual's consumption needs. Although some people may live with family or friends at older ages out of necessity, most would prefer to have financial independence in retirement—something that typically requires individuals to save throughout their prime working years.

Although the economic net worth in the hypothetical economic balance sheet was equal to €680,000, it is possible for an individual to have either a surplus or a shortfall. For example, if the individual is not saving enough to adequately fund the lifestyle he or she will want at retirement, that individual may have a shortfall. Alternatively, if the individual is saving more than enough to fund lifestyle needs and has no bequest goals, he or she may have a surplus. In either case, an economic balance sheet provides some perspective about the overall financial situation of an individual based on his or her holistic wealth.

EXAMPLE 4

Traditional vs. Economic Balance Sheet

1. Contrast a traditional balance sheet with an economic balance sheet.

Solution:

A traditional balance sheet includes assets and liabilities that are generally relatively easy to quantify. An economic balance sheet provides a useful overview of one's total wealth portfolio by supplementing traditional balance sheet assets with human capital and pension wealth and including additional liabilities, such as consumption and bequest goals.

Changes in Economic Net Worth

To provide some context for how the relative value of various household assets changes over a lifetime, we will use the hypothetical example of a British individual at age 25. This 25-year-old is assumed to make £40,000 a year in after-tax income. Over his or her lifetime, real wages are expected to grow at a constant rate of 1% per year, the annual savings rate is 10%, the nominal discount rate is 8%, and the rate of expected inflation is 3%. The value of human capital is estimated using Equation 2. Financial capital at age 25 is assumed to be £10,000, and it is expected to grow at an annual real rate of return of 3% per year. The assumed need from the portfolio is £20,000 for the first year of retirement (age 65) and is increased annually by inflation throughout retirement.

We further assume that at age 30 the individual purchases a home that costs £100,000 in today's currency. The home is purchased with a 10% down payment (which comes from financial capital), with the remainder financed by a 30-year mortgage at

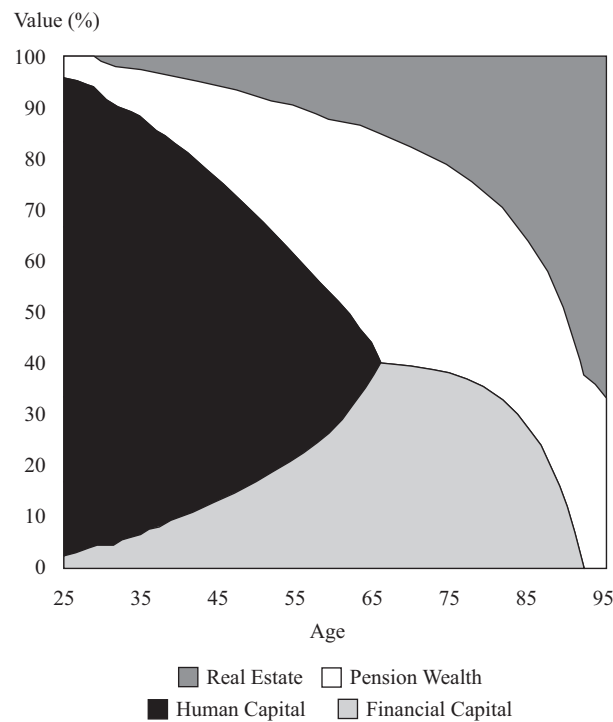
a fixed nominal interest rate of 5%. The real growth rate of the value of the home is assumed to be 1%. Total pension benefits of £20,000 per year (in today's currency, at age 25) are assumed to commence at age 65, and the real discount rate for pension retirement benefits is 5%. We assume the benefits are accrued throughout the employment of the individual.

Exhibit 3 shows the values of the assets in the individual's economic balance sheet and how they are expected to change over time. To simplify the concept, we demonstrate graphically how the sample inputs reflect the allocation in the exhibit.

Exhibit 3: Life-Cycle Economic Balance Sheet Allocation



Traditional balance sheet assets, such as investments in marketable securities, real estate, and businesses, vary in importance from one life-cycle stage to the next. In general, tangible assets, such as real estate and personal goods, which provide great value to a young family, dominate a household's portfolio early in the life cycle. As households age, they accumulate financial assets that must be managed efficiently to provide the greatest expected later-life consumption for the amount of risk the household is willing to take. Non-traditional balance sheet assets, such as employer pensions, increase in importance later in the life cycle, providing an important source of stable consumption and affecting the optimal allocation of securities within an investment portfolio. To illustrate, Exhibit 4 provides the relative weights for the various assets included in Exhibit 3.

Exhibit 4: Relative Weights of Economic Balance Sheet Allocation

In Exhibit 4, we see that for the typical individual, an investment portfolio represents a significant portion of wealth at age 65 but is still less than 50% of the total economic wealth when home equity, pension wealth, and human capital are also considered. As that individual proceeds through his or her retirement years and funds consumption, the relative share of the investment portfolio declines. In the early retirement stage, total economic wealth is dominated by pension wealth (i.e., the remaining mortality-weighted net present value of benefits) and the value of real estate (i.e., the individual's personal residence). For wealthier individuals, the value of defined benefit pension wealth will likely represent a low percentage of the total wealth portfolio in retirement. To the extent that defined benefit pension wealth has very low credit risk (for example, because of the low default probability of National Insurance in the United Kingdom), a retiree's optimal investment portfolio allocation will be affected. As discussed earlier, one must consider the financial health of both the plan and the sponsor providing the defined benefit pension.

In a related manner, a 65-year-old with £2 million in pension wealth will have a higher level of expected remaining lifetime consumption than a retiree with £1 million in pension wealth and the same traditional balance sheet net worth. Both individuals will need to consider means to safeguard the value of the pension wealth as part of the financial planning process. For example, a pension from a private employer may be subject to company-specific risk. The risk of employer insolvency might be hedged in financial markets by positions in securities and derivatives (if available) that have a negative correlation with the value of the company. As mentioned previously, guarantees of benefit payments may exist, such as the Pension Protection Fund in the United Kingdom that protects many private defined benefit schemes.

The allocation of the different asset types will affect the optimal financial asset allocation decision. A 45-year-old individual in Germany with €1 million in human capital and €500,000 in investment assets should invest differently than a 45-year-old

with €3 million in human capital and an identical €500,000 in investment assets. The volatility in the investment portfolio of the individual with lower human capital will have a much greater impact on variation in expected consumption if both individuals have a 40-year planning horizon. Assume that the first 45-year-old with €1.5 million in combined human and financial capital expects to spend approximately €38,000 each year until age 85. The other 45-year-old with €3.5 million in economic net worth expects to spend €88,000 each year. All else being equal, a 40% loss in the first individual's portfolio ($0.4 \times €500,000 = €200,000$) will lead to a 13.2% loss in expected spending per year $[(€200,000/40 \text{ years})/€38,000]$ whereas a 40% investment loss to the second individual's portfolio will lead only to a 5.7% decrease in expected consumption $[(€200,000/40 \text{ years})/€88,000]$. For this reason, portfolio recommendations will be less conservative for the investor with high human capital than for the investor with low human capital if both have the same level of risk tolerance.

EXAMPLE 5

Changes in Human and Financial Capital

1. Describe how the relative values of human capital and financial capital change over an individual's lifetime.

Solution:

The total value of human capital and the total value of financial capital tend to be inversely related over time as individuals attempt to smooth consumption through borrowing, saving, and eventual spending. When human capital becomes depleted, without financial capital, an individual will have no wealth to fund his or her lifestyle. Human capital is generally largest for a younger individual, whereas financial capital is generally largest when an individual first retires.

5

INDIVIDUAL RISK EXPOSURES



discuss risks (earnings, premature death, longevity, property, liability, and health risks) in relation to human and financial capital

Managing risks to financial and human capital is an essential part of the household financial planning process. In this section, we provide an overview of the risks faced by individuals and discuss how they relate to human and financial capital. In future sections, we introduce financial products that could be used to manage many of these risks.

Earnings Risk

Earnings risk, within the context of personal risk management, refers to the risks associated with the earning potential of an individual—that is, events that could negatively affect the individual's human and financial capital. As noted previously, health issues can affect earnings, and some health risks are a function of the occupation itself. For example, a construction worker is likely to face higher health-related

earnings risk than the average worker. Aside from health issues, unemployment and underemployment represent major factors in earnings risk. Sometimes, an employee's job performance or a poor "fit" may lead to job loss, but many people find themselves without a job through no fault of their own.

The risk associated with unemployment for reasons other than disability is rather difficult to characterize. In some cases, such as government employees and union members with seniority, the likelihood of unemployment may be very low. Smaller, younger companies may be riskier employers because of dynamic business conditions or cash flow issues. But even large, well-established companies have been known to go out of business or to close unprofitable divisions or locations. In such cases, even an offer to transfer to another location may be undesirable for someone late in his or her working life, for someone with a working spouse and/or children in school, or for someone who strongly prefers to remain in the same location for a variety of other reasons. Some industries are cyclical and are prone to layoffs, whereas other industries are subject to competitive pressures that may lead to permanent terminations. Self-employed individuals and even some professionals are prone to variability in their earnings. Of course, the cost is the loss or reduction of earnings and may also include the loss of employer contributions to one's retirement fund as well as other benefit programs. A lengthy period of unemployment may itself create more risk because employers are sometimes hesitant to hire people who have been out of work for an extended period of time. If the individual finally finds a job, it may be at a dramatically lower compensation level.

Obviously, the loss of income represents a reduction in both human and financial capital, and this reduction is exacerbated if job opportunities are few, especially in a poor economy or in a region or industry that is particularly affected. For individuals who lose a job as they approach retirement age, it could be very difficult to find another job, even if there are regulations against age discrimination. Aside from the stress on the family budget, unemployment can be psychologically devastating to the individual and his or her family. With earnings risk, as well as health risk (which is discussed in further detail later), an implication in estimating the total value of human capital is that individuals who work in dangerous occupations or in jobs that have a high likelihood of variability or disruption in earnings have either lower future expected earnings or a higher discount rate or both. Financial capital may also be affected by earnings risk because assets will be needed to make up for any loss of income. Furthermore, there may be a need to seek additional training or education to acquire requisite skills, and this retooling can be very expensive.

Premature Death Risk

The term **premature death risk**, which is sometimes referred to as mortality risk, relates to the death of an individual earlier than anticipated whose future earnings, or human capital, were expected to help pay for financial needs and aspirations of the individual's family. These needs include funding day-to-day living expenses, such as food, housing, and transportation, as well as paying off debts, saving for a child's education, and providing for a comfortable retirement for the surviving spouse. An individual's death may also lead to a reduction in the income of the surviving spouse because some family responsibilities of the deceased individual must now be performed by the surviving spouse (assuming the spouse does not remarry). For a young family, the effect can be especially tragic because the increase in household lifestyle that might have accompanied the career of the deceased may never occur (again, if there is no remarriage).

A risk to consumption needs also occurs if a non-earning member of the family dies. The loss can be estimated as the discounted value of the services provided by the deceased family member plus any out-of-pocket death expenses. If a household's

primary caregiver dies, the rest of the family can help with that member's responsibilities, but often additional, paid help is required to replace the primary caregiver's duties. This scenario will mean a dramatic change in lifestyle, compounding the incalculable emotional effect of the death. It could even have a negative impact on the career of the surviving spouse, who may feel drained by the added responsibilities.

Besides the obvious reduction in human capital that the death of an income earner represents, there are also effects on financial capital. Death expenses (including funeral and burial), transition expenses, estate settlement expenses, and the possible need for training or education for the surviving spouse are among the financial costs that may be incurred.

Longevity Risk

Longevity risk within the context of financial planning relates to the uncertainty surrounding how long retirement will last and specifically the risks associated with living to an advanced age in retirement (e.g., age 100). An extended retirement period may deplete the retiree's resources to the point at which income and financial assets are insufficient to meet post-retirement consumption needs. A common question posed to financial planners is, "How much money do I need to have when I retire?" The answer is dependent on the lifespan of the individual, and longevity is a key variable that can only, at best, be estimated. Other important variables include the nominal rate of return on the portfolio, the rate of inflation, additional sources of income (and whether those sources are adjusted for inflation), and the level of spending. Determining how large a fund an individual will actually have at retirement depends on the amount and timing of contributions, the nominal rate of return, and the amount of time until retirement.

When calculating the sum needed at retirement, financial planners often run a Monte Carlo simulation that is based on an assumed asset allocation to calculate the probability that the funds will last for a specified number of years. Another approach for the time variable is to use a mortality table, adjust for health factors, and add years to be conservative. For example, Friedrich is retiring at age 65, and the mortality tables in his country indicate that a 65-year-old man has an expected lifespan of 20 years. But Friedrich is healthy, exercises regularly, eats well, and has had annual physical examinations, and his parents lived until their late 80s, which was past life expectancy at that time. Friedrich might assume that his retirement will last only 20 years (his life expectancy), but the mortality tables indicate a 50% chance that he will live beyond the forecasted period, which is why it is common to add years to be conservative (e.g., plan for retirement to last 30 years, or until age 95). The decision regarding the additional number of years is obviously subjective. The only way to minimize the likelihood of living beyond the forecasted retirement period would be to use extremely advanced ages (e.g., age 110).

Longevity risk can have a significant impact on the lifestyle of an individual. Even in countries that provide significant pension benefits, income may be inadequate to support the hoped-for lifestyle, and insufficient assets may exacerbate the situation. Making matters worse, many pension programs do not consider inflation. Furthermore, some pension programs, even those sponsored by governmental entities, are unlikely to have sufficient assets to pay future expected liabilities without significant changes to the pension structure. Relying on a pension thus entails its own set of risks.

Longevity risk affects human capital in the sense that an individual who is concerned about "living too long" may choose to work longer than someone else might. Indeed, all else being equal, the person who is concerned about outliving his or her money and who intends to work longer has more human capital, but at the possible expense of a less desirable (i.e., longer) retirement stage.

Property Risk

Property risk relates to the possibility that a person's property may be damaged, destroyed, stolen, or lost. There are, of course, many different possible events relating to property risk. A house may catch fire, an automobile may be involved in a collision or be damaged in a hailstorm, or a valuable necklace may be lost. In the context of property risk, *direct loss* refers to the monetary value of the loss associated with the property itself. For example, a house fire may cause €50,000 of damage. If the repair process requires that the family live elsewhere while the damage is repaired, the expenses incurred are considered an *indirect loss*. If the family is renting a room to a boarder, the income lost during construction would also be considered an indirect loss. Similarly, if a driver damages his or her automobile by running into a curb, the damage to the automobile is a direct loss and the cost of renting a replacement automobile is an indirect loss.

Because property represents a financial asset, property risk is normally considered to be associated with a potential loss of financial capital. But property used in a business to create income is rightfully considered in a discussion of human capital. That is, this type of business property can be considered a tool that helps drive future earnings, and to the extent that such property is at risk, human capital is also at risk. Business owners should be especially conscious of the fact that in the absence of insurance or other risk management techniques, both financial and human capital is at risk.

Liability Risk

Liability risk refers to the possibility that an individual or household may be held legally liable for the financial costs associated with property damage or physical injury. In general, one may be *liable* if because of one's action—or inaction when one is legally responsible for taking action—bodily injury, property damage, or other loss is incurred by another person or entity.

For individuals, the most common cause of legal liability involves driving an automobile. An automobile accident may cause bodily injury leading to medical costs, lost income, and even the necessity for long-term care. For the vast majority of people, the potential liability of a major automobile accident exceeds not only their financial capital but also their human capital as well. For example, in some jurisdictions, a liability judgment may result in the confiscation (often termed garnishing) of the wages or other income of the person found liable. Note, though, that the person who is found liable—for example, in an automobile accident—may also have suffered an injury that may affect the individual's financial and/or human capital.

As an example of liability risk, assume that a driver causes an automobile accident in which a passenger in the other car is injured and the other driver's automobile is heavily damaged. In many jurisdictions, the individual who caused the accident is deemed responsible for the repair or replacement of the damaged automobile and the medical expenses and lost income of the victim. As another example, in many countries, a homeowner or even an apartment renter may be deemed legally liable for an accident that causes injury or property damage to a visitor. For example, a guest may accidentally slip on some steps, be seriously injured by the fall, and become incapable of gainful employment. Even if the visitor was careless, laws may specify that the owner or renter of the property is liable.

Health Risk

Health risk refers to the risks and implications associated with illness or injury. Direct costs associated with illness or injury may include coinsurance, copayments, and deductibles associated with diagnostics, treatments, and procedures. In the context

of health insurance, the term *coinsurance* means that the insured must share some of the costs of the specific health care provided. For example, an insurance company may be obligated to pay 80% of the cost of a medical procedure and require that the insured pay the other 20%. *Copayments* refer to the requirement that the insured pay a specified amount of money for a medical service, typically treatment by a physician. For example, a copayment, or “copay,” of US\$30 may be required for a visit to a primary care physician and US\$45 may be required for a visit to a specialist. The remainder of the actual expense is paid by the insurance company. A *deductible* is an amount that the insured is required to spend on health care approved by the insurance company during the plan year before the insurance company pays for anything. For example, there may be a US\$500 deductible per person and a US\$1,500 total deductible for a family. Insurance companies contend that coinsurance, copayments, and deductibles discourage frivolous use of the health care system, thereby keeping insurance premiums lower.

In some countries, health care costs for individuals can be significant. Obviously, the risk associated with these costs varies considerably both across and within countries and must be considered as a risk to financial capital. Health factors typically have a significant impact on the premiums individuals pay for life, disability, and long-term care insurance.

Health risks manifest themselves in different ways over the life cycle and can have significant implications for human capital as well as for financial capital. For example, if a worker becomes disabled as a result of an accident or health incident, he or she may be unable to work while health expenses are incurred, resulting in a loss to both current assets and future earnings. The impact of a negative health event on human capital can be approximated by using the discounted cash flow framework and estimating the decline in projected cash flows along with an increase in the discount rate arising from increased earnings uncertainty. Illness and injury can also obviously have an adverse impact on life expectancy, potentially resulting in death before planned retirement. Furthermore, health issues involving non-earning members of the family can also be costly. There may be a need for special medical services, housing improvements, specialized vehicles, and other health-related expenses. In the case of the special needs of a child, the financial obligation could continue well beyond the parents’ working lives, or even their actual lives.

Although long-term care is a part of the national health care system in some countries, such as Germany and Japan, in others, such as the United States, the cost of long-term care can represent a significant burden on financial capital. In countries where long-term care expenses are incurred by the individual, policies that provide insurance to protect against the cost of long-term care should be considered. Long-term care insurance is designed to cover a portion of the cost of necessities, such as home care, assisted living facilities, and nursing homes.

The risk and cost of long-term care may be considered both a health issue and an issue of insufficient assets at an advanced age—the latter being a component of the aforementioned longevity risk. The risks may also go beyond the immediate family unit. For example, one may have a parent who is not financially capable of paying for long-term care. An added risk is that inflation in long-term care costs (i.e., medical costs) has historically been higher than base inflation.

EXAMPLE 6**Individual Risk Exposures (1)**

1. Describe premature death risk with respect to financial and human capital.

Solution:

Within a personal financial planning context, premature death means that an individual dies before fully providing for his or her financial needs (and, if applicable, those of the family). By definition, at that point human capital is eliminated because the deceased individual can no longer generate income. To a lesser degree, there may also be an impact on financial capital. In addition to expenses associated with a funeral and burial, there may be a need for significant transitional funds or even a requirement to settle certain debts or business obligations upon the individual's death. Funds may also be required for education and/or training of the surviving spouse to generate income.

EXAMPLE 7**Individual Risk Exposures (2)**

1. Describe longevity risk within the context of personal financial planning, and explain how it relates to human and financial capital.

Solution:

Longevity risk refers to the possibility that an individual may live long enough to deplete his or her resources—to outlive one's money. Longevity risk relates primarily to financial capital—that is, spending one's retirement portfolio. But there is also an aspect of human capital in that one may address longevity risk, in part, by retiring later, thus expanding one's retirement portfolio and reducing the number of years to draw it down while increasing one's human capital.

LIFE INSURANCE: USES, TYPES, AND ELEMENTS**6**

- ☐ describe types of insurance relevant to personal financial planning
- ☐ describe the basic elements of a life insurance policy and how insurers price a life insurance policy

An individual's balance sheet provides a comprehensive overview of the asset categories held to fund current and future spending. Each of these categories involves some risk of a random loss. Managing these risks involves assessing possible loss exposures and considering market and non-market solutions to both address the possibility of

and reduce the magnitude of a loss. We review the range of products that can be used to reduce these risks and present a strategy for analyzing the value of possible treatment options.

What are the consequences of risk? Effective risk management for individuals addresses the trade-offs between expected total wealth and security. Individual life-cycle planning involves assessing expected available resources and planning an optimal earning and spending path over a lifetime. But life does not always unfold as expected. A negative event can threaten the value of assets, and a loss in this value will cause total wealth (and expected future consumption) to fall. For each risk exposure, a solution exists to manage that risk exposure, whether through an altered portfolio allocation, a change in behavior, or the purchase of financial and/or insurance products. Each of these solutions involves a cost that generally results in a lower expected level of consumption over time. Shifting assets from risky to risk-free securities results in the loss of a risk premium. In the case of financial products, purchasing insurance trades a reduction in expected lifetime consumption for an increase in the stability of expected spending after a loss. In this section, we discuss the various types of insurance that individuals may use in financial planning. We then turn our attention to annuities, another financial product available to individuals.

Life Insurance

Life insurance protects against the loss of human capital for those who depend on an individual's future earnings. In this section, we provide an overview of the key uses of life insurance, the primary types of life insurance, the basic elements of a life insurance policy, how a life insurance policy is priced, and how to determine the appropriate amount of life insurance to purchase, if any.

Uses of Life Insurance

Life insurance provides a hedge against the risk of the premature death of an earner. A family's need for life insurance is related to the risk of the loss of the future earning power of an individual less the expected future spending of that individual. In each case, the risk associated with premature death can be mitigated by transferring the risk to a third party (i.e., by purchasing life insurance). The optimal amount of insurance to purchase is a function of both the expenses of the insurance hedge and the magnitude of the difference in expected lifetime utility with and without that family member.

Life insurance can also be an important estate-planning tool. A life insurance policy can provide immediate liquidity to a beneficiary without the delay involved in the legal process of settling an estate (i.e., distributing assets to beneficiaries) following the death of an individual. This liquidity can be particularly valuable if the estate contains illiquid assets or assets that are difficult to separate and distribute equitably among heirs.

Another possible use of life insurance is as a tax-sheltered savings instrument, notably in the United States. As mentioned previously in this reading, cash-value policies invest a portion of the premium in a tax-advantaged account that represents the difference between the current cost of providing insurance coverage and the premium. The mortality charge is the cost of providing life insurance, which increases with age (as does mortality risk). As mortality risk increases, the accumulated excess premium can be used to pay the increasingly higher costs of providing insurance protection. These excess premiums can be invested in a variety of instruments that can grow over time sheltered from taxation and can eventually be cashed out without paying for older-age life insurance protection.

Types of Life Insurance

There are two main types of life insurance: temporary and permanent. For the purposes of this reading, both types of life insurance are assumed to be non-cancelable: The policy lapses only at the end of the term (for temporary life insurance) or upon death (for permanent life insurance).

Temporary life insurance provides insurance for a certain period of time specified at purchase. This type of coverage is commonly referred to as “term” life insurance. If the individual survives until the end of the period (e.g., 20 years), the policy will terminate unless it can be automatically renewed. Generally, premiums for term life insurance either remain level over the insured period (e.g., 20 years) or increase over the period as mortality risk increases. The cost of term insurance is less than that of permanent insurance, and the cost per year is less for shorter insured periods (e.g., 10 years versus 20 years), again because of increasing mortality risk.

Permanent life insurance provides lifetime coverage, assuming the premiums are paid over the entire period. Policy premiums for permanent life insurance are usually fixed, and there is generally some underlying cash value associated with a permanent insurance policy. There are several types of permanent life insurance that vary by region. Here, we will discuss the two most common types of permanent life insurance: *whole life insurance* and *universal life insurance*.

Whole life insurance remains in force for an insured's entire life (hence the name). Whole life insurance generally requires regular, ongoing fixed premiums, which are typically paid annually, although monthly, quarterly, and semiannual payment options also exist. Failure to pay premiums can result in the lapse of the insurance policy. There is generally a cash value associated with a whole life insurance policy that may be accessed if the insured chooses to do so. The non-cancelability of whole life insurance can make this type of policy appealing to purchase at younger ages, when an individual is typically healthier. Whole life insurance policies can be participating or non-participating. Participating life insurance policies allow potential growth at a higher rate than the guaranteed value, based on the profits of the insurance company. A non-participating policy is one with fixed values: The benefits will not change based on the profits and experience of the insurance company. Universal life insurance is constructed to provide more flexibility than whole life insurance. The policy owner, generally the insured, has the ability to pay higher or lower premium payments and often has more options for investing the cash value. The insurance will stay in force as long as the premiums paid or the cash value is enough to cover the policy expenses of the provider.

Many permanent life insurance policies have a “non-forfeiture clause,” whereby the policy owner has the option to receive some portion of the benefits if premium payments are missed (i.e., before the policy lapses). The scenarios permitted by a non-forfeiture clause generally include a cash surrender option (whereby the existing cash value is paid out), a reduced paid-up option (whereby the cash value is used to purchase a single-premium whole life insurance policy), and an extended term option (whereby the cash value is used to purchase a term insurance policy, generally with the same face value as the previous policy).

In addition, a number of potential “riders” can be added to both temporary and permanent life insurance policies. Riders are modifications that add some risk mitigation beyond the basic policy. One example of a common rider is an “accidental death” rider (also referred to as accidental death and dismemberment, or AD&D), which increases the payout if the insured dies or becomes dismembered from an accident. Other common riders include an accelerated death benefit (which may allow insured parties who have been diagnosed as terminally ill to collect all or part of the death benefit while they are still alive), guaranteed insurability (which allows the owner to purchase more insurance in the future at certain predefined intervals), and a waiver of premium (whereby future premiums are waived if the insured becomes disabled).

The value of the rider will depend on the level of protection against an unexpected decline in consumption not otherwise provided by a basic policy. An additional way for life insurance policyholders to access the value of the policy is the option to sell the policy to a third party, which is often called a viatical settlement. After purchasing the policy, the third party becomes responsible for paying the premiums and will receive the death benefit when the insured dies.

Basic Elements of a Life Insurance Policy

The basic elements of a life insurance policy include

- the term and type of the policy (e.g., a 20-year temporary insurance policy),
- the amount of benefits (e.g., £100,000),
- limitations under which the death benefit could be withheld (e.g., if death is by suicide within two years of issuance),
- the contestability period (the period during which the insurance company can investigate and deny claims),
- the identity (name, age, gender) of the insured
- the policy owner,
- the beneficiary or beneficiaries,
- the premium schedule (the amount and frequency of premiums due), and
- modifications to coverage in any riders to the policy.

In addition, for a life insurance policy to be valid, the policy owner generally needs to have an insurable interest in the life of the insured. Thus, the presence of an insurable interest is a basic element of an insurance policy as well.

The insured, the policy owner, the beneficiary (or beneficiaries), and the insurer are the four primary parties involved in any life insurance policy. The insured is the individual whose death triggers the insurance payment. The policy owner is the person who owns the life insurance policy and is responsible for paying premiums. The beneficiary is the individual (or entity) who will receive the proceeds from the life insurance policy when the insured passes away. The actual beneficiary of a jointly owned life insurance policy may be determined by the order of death of the prospective beneficiaries (e.g., a husband and a wife). Lastly, the insurer is the insurance company that writes the policy and is responsible for paying the death benefit. The amount payable to the beneficiary is typically referred to as the “face value” of the life insurance policy.

For most life insurance policies, the policy owner and the insured are the same person. In certain instances, however, a policy owner may choose to obtain insurance to protect against a loss in economic value from the death of another individual. For example, as part of a divorce, one ex-spouse may purchase life insurance on the other ex-spouse. Similarly, a business may purchase life insurance on a key executive under the assumption that the business would be negatively affected by that executive’s death.

When the insured is not the policy owner, the policy owner must have an “insurable interest” in the life of the insured. Insurable interest prevents individuals from gambling on the lives of strangers and removes any incentive to hasten the insured person’s demise. An insurable interest means that the policy owner must derive some type of benefit from the continued survival of the individual that would be negatively affected should that individual pass away. For example, a spouse has an insurable interest because he or she relies on the income or household services of the other spouse. A business has an insurable interest in key executives who are essential to the ongoing operations of the business.

Life insurance benefits are payable to the beneficiary upon the death of the insured. Usually, some form of documentation or proof of death is required by the life insurance company, such as a death certificate, before benefits are paid to the beneficiary. Death benefits from a life insurance policy can be paid in various forms, such as a lump sum or an annuity, although lump sums are generally more common.

There may be certain situations in which a life insurance company would not be required to pay a benefit. For example, if the insured commits suicide within some predetermined period after purchasing the policy (e.g., two years), or if the insured made material misrepresentations relating to his or her health and/or financial condition during the application process, benefits may not be payable. There is often a maximum contestability period during which the insurer has a legal right to contest the death benefit, after which the insurer cannot deny the claim even if it involves suicide and/or material misstatement.

EXAMPLE 8

Elements of a Life Insurance Policy

1. Describe the concept of insurable interest for life insurance.

Solution:

An insurable interest means that the policy owner must derive some type of benefit from the continued survival of the insured that would be negatively affected should the insured pass away. For example, an individual may rely on a spouse for his or her financial well-being. If the spouse dies, income is no longer generated, leading to financial problems. Another example is a business that may have an insurable interest in a key employee who generates large sales volumes. The purpose of an insurable interest is to prevent individuals from gambling on the lives of others or from having a financial reason to arrange the death of the insured.

LIFE INSURANCE PRICING, POLICY COST, AND AMOUNT NEEDED

7



describe the basic elements of a life insurance policy and how insurers price a life insurance policy

There are a number of factors that determine how an insurer prices life insurance, and there are many different types of life insurance policies. Although the details of the actuarial calculations are beyond the scope of this reading, it is useful to understand the basic concepts of life insurance pricing.

In general, there are three key considerations in the pricing of life insurance: mortality expectations, a discount rate, and loading.

Mortality Expectations

One of the most important factors in determining the price for life insurance is the expected mortality of the insured individual (i.e., how long the person is expected to live). Actuaries at insurance companies estimate mortality based on both historical data and future mortality expectations. Generally speaking, life expectancies in most regions of the world have been increasing. Certain attributes, such as age and gender, are obvious factors in evaluating life expectancy. Whether the applicant is a smoker (or has other health risks) is another important factor because smoking is associated with deadly diseases. Exhibit 5 shows an example of the probability of men and women (both smokers and non-smokers) dying at various ages, although these numbers will vary considerably in different countries.

Rather than use a generalized mortality table, life insurance company actuaries typically make adjustments to consider additional factors. The underwriting process serves to categorize applicants according to their perceived riskiness, consistent with the actuaries' specifications. The resulting customized tables consider applicants' health history, particularly conditions that are associated with shorter-than-average life expectancy, such as cancer and heart disease. If an applicant's parents or siblings died at a relatively early age from certain diseases, that applicant may be considered a bigger risk. Excess weight is another health issue leading to shorter life expectancies. Certain activities, such as scuba diving and flying personal aircraft, are deemed to increase mortality risk also. All of these underwriting factors can be collected on a typical life insurance application, and the salesperson who gathers this information can be considered the first level of the underwriting process.

For larger policies, insurance companies may require a physical examination, performed by an insurer-paid nurse or physician, and the examination could include blood pressure, cholesterol and other blood analysis, an electrocardiogram, and other tests. All of these factors can be used to categorize applicants in tables that discriminate among standard risks, preferred (lower) risks, and high risks, and the cost to the insured can vary considerably. Of course, some people have a sufficient number of factors, or serious-enough factors, to make them uninsurable. This underwriting process reduces the likelihood of *adverse selection*. Adverse selection refers to the fact that individuals who know that they have higher-than-average risk are more likely to apply for life insurance. Unless the insurance company performs its underwriting well, mortality experience can be worse than projected.

THE PROBABILITY OF DYING AT CERTAIN AGES

Exhibit 5 provides information about the mortality (i.e., the probability of dying) for males and females at different ages.² The cost of life insurance is based on the probability that the insured will die during the duration of the policy. The table helps demonstrate why younger (versus older) individuals, females (versus males), and non-smokers (versus smokers) tend to pay less for life insurance—the expected probability of dying in a given year is lower.

² Data are based on the American Academy of Actuaries' 2017 Commissioners Standard Ordinary (CSO) Tables. <https://www.soa.org/experience-studies/2015/2017-cso-tables/> accessed 21 November 2018

Exhibit 5: Mortality of Males and Females at Certain Ages

Age	Male			Female		
	Composite	Non-Smoker	Smoker	Composite	Non-Smoker	Smoker
35	0.14%	0.09%	0.14%	0.08%	0.07%	0.10%
40	0.21%	0.15%	0.24%	0.12%	0.10%	0.17%
45	0.26%	0.19%	0.35%	0.14%	0.11%	0.23%
50	0.30%	0.23%	0.48%	0.21%	0.15%	0.37%
55	0.42%	0.35%	0.74%	0.32%	0.25%	0.60%
60	0.67%	0.50%	1.21%	0.52%	0.37%	1.00%
65	1.12%	0.84%	2.08%	0.88%	0.59%	1.66%
70	1.81%	1.40%	3.35%	1.48%	0.95%	2.61%
75	3.18%	2.58%	5.34%	2.45%	1.71%	3.93%
80	5.38%	4.65%	7.56%	4.23%	3.33%	6.27%
85	9.71%	8.80%	11.75%	7.77%	6.54%	10.74%
90	17.41%	16.55%	19.04%	13.79%	12.27%	17.34%
95	25.49%	25.16%	26.09%	21.96%	20.82%	24.65%

EXAMPLE 9**Mortality Expectations**

1. If a given male and female are the same age and have equivalent health profiles, evaluate which one should expect to pay more for life insurance.

Solution:

A key pricing component of life insurance is expected mortality. From Exhibit 5, one can see that the chance of death for females across the age spectrum is less than it is for males of the same age. Therefore, all else being equal, females should expect to pay less than males for the equivalent life insurance.

Calculation of the Net Premium and Gross Premium

The *net premium* of a life insurance policy represents the discounted value of the future death benefit. To illustrate a simplified calculation of the net premium, we will consider the example of a one-year, non-renewable term life insurance policy with a death benefit of US\$100,000 for Ramon, a 40-year-old non-smoking male. The insurance company insures thousands of people with characteristics like Ramon's. Thus, the life insurance company will experience a predictable distribution of death benefit payments in a given year, although it does not know who among its customers will die during that year.

Premiums are collected at the beginning of the year, and for simplicity, we will assume that death benefit payments occur at the end of the year. As shown in Exhibit 5, and in the absence of other underwriting information, Ramon has a probability of

0.15% of dying within the year. Although the life insurance company will pay a death benefit of either US\$100,000 or US\$0, we can calculate an expected outflow at the end of the year of US\$150, which equals $(0.0015 \times \text{US\$100,000}) + (0.9985 \times \text{US\$0})$. Finally, a discount rate, or interest factor, representing an assumption of the insurance company's return on its portfolio, is applied to the expected outflow. Assuming a 5.5% rate, US\$150 is discounted by one year to a present value of US\$142.18 ($\text{US\$150}/1.055$), which is the net premium.

As mentioned previously, life insurance companies typically offer level term policies, under which the insured can pay equal annual premiums for a specified number of years—for example, a five-year level term policy. The calculation still requires discounting expected future death benefit payments back to the present, but we must also consider the fact that the individuals who die within the five-year period will not be paying premiums for the remaining outstanding term.

To determine what the insurance company would actually charge Ramon for the one-year policy, the insurer must consider other factors to calculate the *gross premium*. The gross premium adds a *load* to the net premium, allowing for expenses and a projected profit for the insurance company.

Expenses are incurred by the insurer for both writing a life insurance policy and managing it on an ongoing basis. Expenses associated with writing a life insurance policy include the costs of the underwriting process, which potentially include a sales commission to the agent who sold the policy and the cost of a physical exam. Ongoing expenses include overhead and administrative expenses associated with monitoring the policy, ensuring that premiums are paid on a timely basis, and verifying a potential death claim. Furthermore, most companies provide a low percentage “renewal commission” for the first years of the policy, which encourages the agent to provide needed advice to the policy owner and to try to keep the policy owner from terminating the policy.

Life insurers can be divided into two groups—stock companies and mutual companies. Stock companies are similar to other corporations in that they are owned by shareholders, have a profit motive, and are expected to provide a return to those shareholders. Within the constraints of supply and demand for their product, stock life insurance companies add a projected profit as a part of the load in pricing their policies. In contrast, mutual companies are owned by the policy owners themselves and there is no profit motive. Mutual companies typically charge a gross premium that is somewhat higher than the net premium plus expenses, even though mutual companies do not have profits per se. Then, if mortality experience, expenses, and/or investment returns are better than projected, the amount by which the gross premium exceeds the net premium plus expenses may be paid back to the policy owners as a policy dividend, which is considered a return of premium to the policy owner rather than income.

Premiums for level term policies are higher than those for annually renewable (one-year) policies in the early years. But premiums are lower in the later years of the policies—most notably for longer periods, such as a 20-year level term—because annually renewable term policies often have rapidly increasing premiums. As can be seen in Exhibit 5 earlier in this section, mortality rates begin accelerating rather quickly after age 40. Life insurers sometimes offer low initial rates on annually renewable policies with the expectation that many purchasers of these policies will simply pay the increasing premiums.

Some consumers buy an annually renewable term policy with the intention of taking advantage of the “loss-leader” pricing in the early years and then, when rates rise too much, switching to another company that has a lower premium at the newly attained age. Unfortunately, there is risk in this strategy in that a health issue or accident could make that individual uninsurable, leaving him or her with an annually renewable policy that has an escalating premium.

EXAMPLE 10**Life Insurance Pricing**

1. Discuss the three most relevant elements of life insurance pricing.

Solution:

The three most relevant considerations in pricing life insurance are mortality expectations, the discount rate, and loading.

- **Mortality expectations:** The insurer is concerned about the probability that the insured will die within the term of the policy. Actuaries evaluate mortality expectations based on historical experience, considering such factors as age and gender, the longevity of parents, blood pressure, cholesterol, whether the insured is a smoker, and whether the insured has had any diseases or injuries that are likely to lead to death during the policy term.
- **Discount rate:** A discount rate, or interest factor, representing an assumption of the insurance company's return on its portfolio, is applied to the expected outflow.
- **Loading:** After calculating the net premium for a policy, which may be considered the pure price of the insurance, the insurance company adjusts the premium upward to allow for expenses and profit. This adjustment is the load, and the process is called loading.

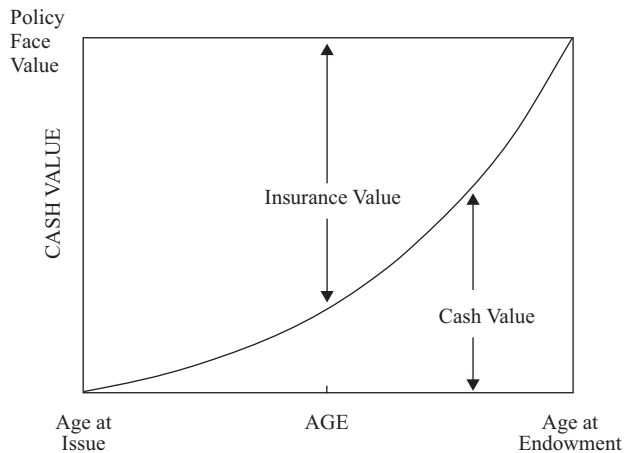
Cash Values and Policy Reserves

As noted earlier, although initial premiums are higher, whole life policies offer the advantage of level premiums and an accumulation of cash value within the policy that (1) can be withdrawn by the policy owner when the policy endows (or matures) or when he or she terminates the policy or (2) can be borrowed as a loan while keeping the policy in force. These cash values build up very slowly in the early years, during which the company is making up for its expenses. For example, just the first-year commission on a whole life policy could be equivalent to 100% of the first-year premium; thus, the company is "in the hole" for the first few years and trying to recover the initial expenses. The commissions decline in subsequent early years as the effort required by a sales agent to service the policy is reduced.

Exhibit 6 shows a representation of the build-up of cash value within a whole life policy that endows at a specified age, perhaps age 100. It is important to recognize the interrelationship of the following amounts:

- The premium stays constant.
- The face value stays constant.
- The cash value increases.
- The insurance value decreases.

Essentially, as cash values increase and the insurance value decreases, the ongoing premium is paying for less and less life insurance.

Exhibit 6: Build-up of Cash Value in a Whole Life Insurance Policy

To the extent that the life insurance is intended to replace human capital, it may become unnecessary after the individual's working years are finished. Enough financial capital may have been accrued to make even the immediate death expenses payable from other funds without the need for life insurance. The existence of increasing cash values within a policy adds another dimension to the decision whether to terminate a policy.

Note also that life insurers are typically required by regulators to maintain *policy reserves*, which are a liability on the insurance company's balance sheet. Policy reserves become especially important for whole life policies. With a whole life policy, the insurance company specifies an age at which the policy's face value will be paid as an endowment to the policy owner if the insured person has not died by that time. The insurance company must accumulate reserves during the life of the policy to be able to make that payment. The policy reserve can be defined as follows: Policy reserve = Present value of future benefits – Present value of future net premiums. From the equation, we can infer that as the insured person gets older, the present value of the future death benefit (or the cash value that could be withdrawn) gets larger. At the same time, the present value of the future net premiums gets smaller because fewer premiums remain. Thus, the policy reserves must grow larger over time until, at the time of endowment, the reserves equal the present value of the future benefits. (See Rejda and McNamara 2014, p. 272.)

Consumer Comparisons of Life Insurance Costs

It may seem easy to compare the cost of two different policies of equal size by simply looking at the first year's premiums. Although this may be the case with term policies, comparing the cost of two whole life insurance policies is much more complex because, for example, one policy may have larger premiums but also faster growth of cash values.

Consumer and/or governmental organizations may provide comparisons of policies from various life insurance companies based on projected data, including assumptions about future dividends and cash values. In many jurisdictions, regulations require that life insurance companies provide consumers with cost data that consider the time value of money and assume a specified number of years. The two most popular indexes for comparison are the *net payment cost index* and the *surrender cost index*,

both of which calculate a cost per year per thousand dollars of life insurance coverage under different sets of assumptions. Both methods assume a specific time period, such as 20 years, and a specific compounding discount rate, such as 5%.

The net payment cost index assumes that the insured person will die at the end of a specified period, such as 20 years. Calculation of the net payment cost index includes the following steps:

- A.** Calculate the future value of an annuity due of an amount equal to the premium, compounded at a 5% discount rate for 20 years. An annuity due—an annuity for which the premium payment is received at the beginning of the period (versus an ordinary annuity, for which the premium payment is received at the end of the period)—is used because premiums are paid at the beginning of the period.
- B.** Calculate the future value of an ordinary annuity of an amount equal to the projected annual dividend (if any), compounded at 5% for 20 years. An ordinary annuity is used because dividend payments are made at the end of the period.
- C.** Subtract B from A to get the 20-year insurance cost.
- D.** Calculate the payments for a 20-year annuity due with a future value equal to C and a discount rate of 5%. This amount is the interest-adjusted cost per year. Again, an annuity due is used because premium payments occur at the beginning of the year.
- E.** Divide by the number of thousand dollars of face value.

The surrender cost index assumes that the policy will be surrendered at the end of the period and that the policy owner will receive the projected cash value. Calculation of the surrender cost index includes the following steps:

- A.** Calculate the future value of an annuity due of an amount equal to the premium, compounded at 5% for 20 years. We use an annuity due here for the same reason indicated for the net payment cost index.
- B.** Calculate the future value of an ordinary annuity of an amount equal to the projected annual dividend (if any), compounded at 5% for 20 years. We use an ordinary annuity here for the same reason indicated for the net payment cost index.
- C.** Subtract B and the Year 20 projected cash value from A to get the 20-year insurance cost.
- D.** Calculate the payments for a 20-year annuity due with a future value equal to C and a discount rate of 5%. This amount is the interest-adjusted cost per year.
- E.** Divide by the number of thousand dollars of face value.

For example, a US\$100,000 face value whole life policy has an annual premium of US\$2,000, paid at the beginning of the year. Policy dividends of US\$500 per year are anticipated, payable at year-end. A cash value of US\$22,500 is projected for the end of Year 20.

Net Payment Cost Index Calculation

Future value of premiums (annuity due): US\$2,000 annual payment, 20 years, 5%	US\$69,439
Future value of dividends (ordinary annuity): US\$500 annual payment, 20 years, 5%	-16,533
20-year insurance cost	US\$52,906
Annual payments for 20-year insurance cost (annuity due): 20 years, 5%	1,524
Divide by US\$ thousands of face value	÷ 100
Net Payment Cost Index, cost per US\$ thousand per year	US\$15.24

Surrender Cost Index Calculation

Future value of premiums (annuity due): US\$2,000 annual payment, 20 years, 5%	US\$69,439
Future value of dividends (ordinary annuity): US\$500 annual payment, 20 years, 5%	-16,533
20-year cash value (given above)	-22,500
20-year insurance cost	US\$30,406
Annual payments for 20-year insurance cost (annuity due): 20 years, 5%	876
Divide by US\$ thousands of face value	÷ 100
Surrender Cost Index, cost per US\$ thousand per year	US\$8.76

The major benefit of these indexes is the ease of comparing policies of the same type. Generally speaking, the lower the index value is, the better the value. However, policies do not always perform as projected. If the insurance company's actual return, expense, and/or mortality experiences are worse than projected, the *ex post* indexes could be larger.

CALCULATING LIFE INSURANCE NEEDS

Two distinctly different methods are commonly used to calculate the amount of life insurance needed. The *human life value* method is consistent with the concept of human capital. It involves replacing the estimated net contribution to family finances that the insured would generate if that individual did not die during his or her projected earning life. In general, this calculation involves estimating future income that would be generated by the insured, offset by incremental expenses that would be attributable to the insured. The net amounts in each year are then discounted back to the present to calculate the amount of insurance needed. An amount may be added to cover so-called "final expenses," such as funeral and other death expenses.

The *needs analysis* method, as the name implies, is concerned with meeting the financial needs of the family. Needs analysis typically involves estimating living expenses for survivors for an appropriate amount of time, typically until adulthood for surviving children and to projected life expectancy for a surviving spouse. Also included are education costs, final expenses, and any other special expenses. These amounts are discounted back to the present to calculate the total funds needed. Any assets available are subtracted, and the amount remaining is the life insurance needed.

Both the human life value and needs analysis methods are demonstrated in detail in Section 7.5.

How Much Life Insurance Does One Need?

The optimal amount of life insurance for an individual will vary based on a number of factors. Some individuals with no dependents or bequest goals may not need any insurance, whereas an individual with young children and a non-working spouse may need a significant amount of life insurance. In this section, we outline some of the key considerations to use when determining how much life insurance to purchase.

The primary purpose of life insurance is to replace the present value of future earnings. A 65-year-old individual without children will experience an emotional loss from the premature death of a spouse, as well as some additional short-term expenses, but the economic loss will be modest (or even negative if expected spending needs fall in retirement). Other reasons to consider life insurance include the following:

- *Immediate financial expenses:* These include direct costs associated with death, such as funeral and legal expenses. Although the costs associated vary by region, funerals can be expensive and represent a potentially sizable financial burden on the family. Additional immediate financial expenses could include covering the short-term loss of wages.
- *Legacy goals:* In addition to income replacement, an individual may use life insurance to achieve certain legacy goals. These can include gifts to charities, bequests to family members, and estate planning.

To calculate the amount of insurance needed, one should estimate the amount of money needed to restore the present value of expected earnings that would have occurred if the earner remained alive. Because the purpose of life insurance is to smooth consumption by preventing a drop in spending from an unexpected premature death, the value of life insurance should be equal to the difference in household spending with and without the human capital of the earner. An accurate calculation will adjust household spending needs downward to compensate for the reduction in household size, in addition to estimating the present value of the expected human capital of the insured. The cost of the insurance is obviously an important consideration as well.

Another important consideration when purchasing life insurance is the insurance company's ability to meet its financial obligations. Company financial strength is evaluated by various rating agencies and is important because it provides an indication of the ability of the insurer to meet its obligations and to weather adverse market conditions.

EXAMPLE 11

Appropriateness of Life Insurance

1. Consider two potential life insurance candidates: (1) a 40-year-old doctor who is married with two young children, substantial student loans, and sizable earnings; and (2) a 35-year-old single person with a moderate amount of financial wealth. Based on the information presented, which person would be a more appropriate candidate for life insurance and why?

Solution:

The first individual is a much more appropriate candidate for life insurance. Given that the doctor has substantial debt and high earnings, the value of this individual's earnings potential (human capital) is significant. Likewise, with two young children, there is a high dependence on future earnings, representing another reason the earnings potential should be hedged with life insurance. In contrast, the 35-year-old does not have any beneficiaries

that would need to be supported. But the younger individual may want to consider purchasing insurance while he or she is still insurable.

8

OTHER TYPES OF INSURANCE

- ☐ describe types of insurance relevant to personal financial planning

Disability income insurance is designed to mitigate earnings risk as a result of a disability, which refers to the risk that an individual becomes less than fully employed because of a physical injury, disease, or other impairment. Many disabilities for gainfully employed individuals are relatively short rather than lifelong, but the financial disaster associated with the possibility of a lifelong disability can be addressed in a comprehensive risk management plan.

What is meant by disability? The definition of disability used by insurance providers typically specifies one of the following:

- Inability to perform the important duties of one's regular occupation
- Inability to perform the important duties of any occupation for which one is suited by education and experience
- Inability to perform the duties of any occupation

Consider a surgeon who loses the use of his or her dominant hand in an accident. By the first definition, this individual would likely be deemed fully disabled. By the second definition, this individual might be able to work as a general practitioner physician. By the third definition above, as long as the individual was able to be employed (e.g., as a professor at a medical school), he or she might not be considered disabled. For most people, especially professionals with specialized skills, policies that use the first definition are best, even though they are more expensive. In general, a disability income insurance policy will specify that some percentage of the difference between the pre-injury income and the post-injury income would be paid to the insured.

As with all insurance policies, disability income insurance policy standards vary widely in different jurisdictions and even for different companies. For most policies, the premium is fixed and based on the age of the insured at the time of policy issue, and the policies are underwritten for the health and occupation of the insured. Disability income coverage is available both through individual policies and through many employers. Disability income policies usually include provisions for partial and residual disability. Partial disability means that although the insured cannot perform all the duties of his or her occupation, the individual can perform enough to remain employed, albeit at a lower income. Partial disability provisions pay a reduced benefit, providing a financial incentive for the insured to get back to work as soon as possible. Residual disability refers to the possibility that although the insured can perform all the duties of his or her occupation, the individual cannot earn as much money as before. Consider the surgeon who can still perform operations but because of a back injury cannot perform as many surgeries per day as before the injury. The reduction in income would be addressed by the residual disability benefit, which is smaller than the benefit for full disability.

Typically, insurers will cover compensation only up to specific amounts. Furthermore, they will insure only up to a specific percentage of compensation, perhaps 60%–80%, for two reasons. First, if the insured becomes disabled, other expenses decrease, such

as certain payroll taxes, commuting costs, clothing, and food; thus, full replacement is not necessary. Second, there is a greater chance for fraudulent claims if the disability income payments are close to the normal compensation.

Other aspects of disability income insurance include the following:

- The *benefit period* specifies how long payments will be made. A specified number of years may be stated in the policy, but typically, the benefit period lasts until normal retirement age, which varies globally between approximately 55 and 70. The age limit discourages the filing of fraudulent claims intended to substitute disability benefits for retirement income. Usually, a minimum number of years of benefits, perhaps five, is specified in case the individual should become disabled within that many years of the specified age. For example, a 62-year-old who becomes disabled would receive benefits to age 67. This provision encourages individuals to maintain their policies all the way to retirement.
- The *elimination period*, or *waiting period*, specifies the number of days the insured must be disabled before payments begin being made. Naturally, the shorter the elimination period, the higher the premiums. A typical elimination period for policies in the United States is 90 days.
- The *rehabilitation clause* provides payments for physical therapy and related services to help the disabled person rejoin the workforce as soon as possible.
- The *waiver of premium* clause specifies that premiums need not be paid if the insured becomes disabled, and it often includes a reimbursement of premiums during the elimination period.
- The *option to purchase additional insurance rider* allows the insured to increase coverage without further proof of insurability, albeit at the rate appropriate for the insured's current age.
- A *non-cancelable and guaranteed renewable* policy guarantees that the policy will be renewed annually as long as premiums are paid and that there will be no changes to premiums or promised disability benefits until, usually, age 65. Even if employment income declines during the working life, the monthly benefit will remain at the level specified in the policy.
- A *non-cancelable* policy cannot be canceled as long as premiums are paid, but the insurer can increase premiums for the entire underwriting class that includes the insured. This version is less expensive than a non-cancelable and guaranteed renewable policy, but one should be aware that insurance companies with significant loss experience are likely to raise rates.
- Inflation adjustments to benefits may be provided by a *cost of living rider*, which will adjust benefits with an accepted index or by a specified percentage per year.

Note that, as with virtually all insurance types, any provision in a disability insurance policy that appears advantageous to the insured party will likely increase the premium.

Property Insurance

Property insurance is used by individuals to manage property risk, which was discussed earlier in the reading. Although property insurance coverage applies to a multitude of situations, for most individuals, the primary areas to cover are the home/residence and the automobile.

Homeowner's Insurance

With regard to a residence, homeowner's insurance is designed to address risks associated with home ownership as well as risks associated with personal property and liability. Renter's insurance is similar to homeowner's insurance but without coverage on the structure. Property insurance protects the insured in case of loss related to his or her property. As we discussed earlier, there are, of course, many different possible events relating to property risk, such as a house fire, an automobile collision, a stolen television, or a lost necklace.

Homeowner's policies may be specified as "all-risks," which means that all risks are included except those specified, or as "named-risks," which means that only those risks specifically listed are covered. All-risks policies are generally more costly. Homeowner's insurance may also be available in either of two versions, based on the way a claim is settled. A policy based on *replacement cost* will reimburse the insured person for the amount required to repair a damaged item or replace a lost, destroyed, or stolen item with a new item of similar quality at current prices. A policy based on *actual cash value* will reimburse the insured person for the replacement cost less depreciation. The replacement cost version is a more expensive policy.

As mentioned earlier in the reading, a *deductible* is the amount of a loss that must be absorbed by the policy owner before the insurance company will make any payment. Deductibles represent a form of active risk retention. If the homeowner's policy has a US\$1,000 deductible and there is hail damage of \$10,000, the homeowner must pay the first \$1,000 and the insurance company would be liable for the remaining \$9,000. Deductibles ensure that the homeowner retains some responsibility (and risk) associated with a loss.

As part of their business models, insurance companies price their policies to encourage the use of higher deductibles. For the consumer, this means that a cost-benefit analysis should be performed when determining the optimal deductible level because a larger deductible likely means a lower insurance premium. Imagine that a policy with a \$1,000 deductible for a given property has an annual premium of \$2,000 and a policy with a \$500 deductible has an annual premium of \$2,100. The consumer should recognize that selecting the second policy means essentially paying \$100 more for a \$1,000 insurance policy that has a \$500 deductible. An alternative way of looking at it is to ask whether it is worth \$100 to insure the second \$500 of loss.

Some individuals underinsure their homes to save money, but they do so at their own risk. If a potential loss could exceed the amount of insurance, that individual is retaining risk of that excess amount, either consciously or unconsciously. For individuals who have significant wealth, whose home is a relatively small percentage of their net worth, and who have adequate liquidity, it may make sense to self-insure these types of risks (i.e., maintain only a limited amount of homeowner's insurance or own a policy with a very high deductible).

Mortgage lenders typically require that the homeowner carry enough insurance that if the mortgagor dies, a total loss would trigger payment of an amount at least equal to the outstanding mortgage. Because mortgage balances typically decline over time, insurance contracts covering the outstanding mortgage can be purchased with a decreasing face value and decreasing premiums.

Insurance companies have a different interest from that of mortgage lenders. They want the house to be insured for its full value—less the value of the land because the land will not be destroyed—or at least a high percentage of full value. Premiums are calculated based on this assumption. Although the insurance company is obligated to pay only the face value of the policy in the case of a total loss, it is at a disadvantage if partial losses occur.

Consider a \$500,000 (replacement cost) home that, because inflation in home prices has been ignored, is insured for only \$250,000. In the absence of other contractual restrictions, if the house sustains \$250,000 of damage in a fire, the insurance company

would have to pay the entire face value of the policy even though the house sustained only a 50% loss. From the company's standpoint, it should have been receiving the larger premiums for a \$500,000 house to pay \$250,000 for a 50% loss.

To offset this dilemma, losses are reimbursed at a lower rate if the home is underinsured. It is common for an insurance company to reduce payments if the home is insured for less than 80% of its replacement cost.

Homeowners' liability risks are typically addressed within the insurance policy on the home. That is, there is a provision in the policy for liability coverage for a specified amount in case, for example, a visitor is injured in an accident at the home. This coverage excludes professional liability, such as physicians' malpractice insurance, and business liability, which should be covered with separate policies. The homeowner's policy also excludes liability resulting from intentional acts—for example, throwing a chair through a neighbor's window.

Aside from purchasing homeowner's insurance, one can address homeowner risk through other risk management techniques. The following are some examples:

- Risk of theft of valuable financial documents can be avoided by storing them in a bank's safe deposit box.
- Risk of overall theft can be reduced through the use of high-quality locks, alarms, and surveillance systems.
- Risk of loss or corruption of electronic data can be avoided by storing backups offsite.
- Risk of damage to electronic equipment from a power surge can be reduced by installing surge protectors.
- Risk of loss from fire can be reduced through the use of fire-resistant building materials—and through the easy availability of fire extinguishers.

Automobile Insurance

Automobile property risk can also be addressed through various risk management methods. For example, one might avoid driving in inclement weather conditions. One might require passengers to wear seat belts, reducing their likelihood of injury. One might use common, frequently promoted safe driving techniques or even take alternate routes that are less risky. If buying a new car, one might consider a vehicle with a backup camera and lane-change warning system.

Automobile insurance rates are primarily based on the value of the automobile and the primary operator's age and driving record. Other factors are also included but vary considerably among jurisdictions. Coverages for damage to the automobile are typically divided into two parts. *Collision coverage* is for damage from an accident, and *comprehensive coverage* is for damage from other sources, such as glass breakage, hail, and theft. There may also be coverage available in case one's automobile is damaged by an uninsured or underinsured driver, as well as medical coverage for passengers in the insured's automobile. Insurance companies normally insure automobiles only up to the cost of replacing the automobile with one of the same make and model and in the same condition. If the cost to repair the automobile exceeds its actual cash value, the insurance company typically reimburses only the amount of the actual cash value.

As with homeowner's insurance, automobile owners typically retain some risk through the use of deductibles or by avoiding collision and comprehensive coverage. Again, selecting the amount of the deductible or rejecting property coverage involves a simple cost-benefit analysis. If the individual is able to bear the wealth risk and to afford repairs or a new car if damage occurs, then risk retention will increase expected wealth over time.

Liability associated with automobiles is typically covered under the same policy, with specified limits for bodily injury and property damage. In most countries, some level of third-party coverage is mandatory and additional insurance may be purchased to increase the level of protection. Liability limits often vary for different types of loss—for example, higher limits to cover the costs of physical injury and separate limits to cover the loss of property. If actual liability in an accident exceeds these amounts, the automobile owner is responsible for the remainder.

Health/Medical Insurance

Any discussion of **health insurance** is highly dependent on the country of residence. In certain countries, health care is governmentally funded and there is no private health insurance. In others, there is a two-tiered system, with governmental coverage for everyone and upgraded coverage for additional payments.

In the United States, one type of insurance approach is called an *indemnity plan*, which allows the insured to go to essentially any medical service provider, but the insured must pay a specified percentage of the “reasonable and customary” fees. Another type of plan is a *preferred provider organization* (PPO), which is a large network of physicians and other medical service providers that charge lower prices to individuals within the plan than to individuals who obtain care on their own. A third type of plan is a *health maintenance organization* (HMO), which allows office visits at no, or very little, cost to encourage individuals to seek help for small medical problems before they become more serious.

Comprehensive major medical insurance covers the vast majority of health care expenses, such as physicians’ fees, surgical fees, hospitalization, laboratory fees, x-rays, magnetic resonance imaging (MRIs), and most other expenses that are “reasonable and customary” and part of generally accepted medical care. Aside from the premiums for the actual coverage, major medical insurance includes several other provisions that can substantially influence the total financial outlay. Below are some of the key terms of most health (medical) insurance plans:

- *Deductibles* refer to the amount of health care expenses that the insured person must pay in a year before any expense reimbursement is paid by the insurance company.
- *Coinsurance* specifies the percentage of any expense that the insurance company will pay, often 80%, with the insured person responsible for the remainder.
- *Copayments* are fixed payments that the insured must make for a particular service, such as a doctor’s office visit.
- *Maximum out-of-pocket expense* refers to the total amount of expenses incurred within a year beyond which the insurance company pays 100%. It is often expressed in terms of an individual maximum and a family maximum. This concept is often referred to as a stop-loss limit.
- *Maximum yearly benefit* refers to the maximum amount that the insurance company will pay in a year.
- *Maximum lifetime benefit* refers to the maximum amount that the insurance company will pay over an individual’s lifetime.
- *Preexisting conditions* refer to health conditions that the insured had when applying for insurance. They may or may not be covered by the insurance company, depending on the policy, laws, and regulations.
- *Preadmission certification* refers to a requirement that the insured receive approval from the insurer before a scheduled (non-emergency) hospital stay or treatment.

Of course, the cost of a plan will be affected by the degree of inclusion and specified value of the preceding provisions. Besides the cost of the plan, health insurance purchasers should be concerned about the breadth and quality of the network of physicians and hospitals available to insured individuals.

Liability Insurance

To manage liability risk, individuals often obtain **liability insurance**. It is possible that the amount of liability coverage in the homeowner's and automobile insurance policies is less than one thinks is appropriate. In that case, it is reasonable to purchase a *personalumbrella liability insurance* policy. This type of policy has specified limits but pays claims only if the liability limit of the homeowner's or automobile policy is exceeded. For example, consider an individual whose automobile policy specifies a property damage liability limit of US\$100,000 and who has an umbrella policy with a liability limit of US\$1 million. If that individual is responsible for an automobile accident that causes US\$300,000 of damage, the automobile policy would pay the first US\$100,000 and the umbrella policy would pay the remaining US\$200,000. Umbrella policies are relatively inexpensive.

It is common for people to think of liability insurance coverage in terms of protecting one's own financial assets. Some jurisdictions may specify exempt property—one's home or retirement savings, for example—that cannot be seized in the case that liability for an accident exceeds the amount of insurance coverage. Thus, one might ask oneself, "What is the worst thing that could happen to me in case of an extremely large liability judgment?" But if the goal is to indemnify the injured party—that is, to make the injured party "whole" financially—the amount of insurance purchased should be based on the potential financial catastrophe that could face an injured party.

Other Types of Insurance

Other types of risks may be present, depending on the individual's situation. For example, when purchasing a home, many individuals purchase *title insurance* (and in some jurisdictions, it is required). The purpose of title insurance is to make sure that ownership of the property is not in doubt. Personal watercraft and trailers may require a separate insurance policy or an endorsement, which is a form of insurance coverage added to an existing policy to cover risks that are not otherwise included in that policy. Again, appropriate levels of risk avoidance, reduction, prevention, and retention will depend on the situation and the size and probability of the potential loss.

One may consider service contracts when purchasing an automobile, home appliance, or other sizable product to avoid repair costs. These types of pseudo-insurance are profit centers for the companies selling them and are often relatively expensive for consumers, partly because of a lack of competition. For example, at the time of purchase of an automobile, one may be offered the opportunity to purchase an extended warranty at the same time from the automobile dealer. There is limited opportunity for price shopping, and the price of the service contract can usually be rolled into the automobile loan. Under these circumstances, the automobile dealer can charge a high rate to the disadvantage of the purchaser. Note that even this type of contract typically involves a deductible.

9

ANNUITIES: TYPES, STRUCTURE, AND CLASSIFICATION



discuss the use of annuities in personal financial planning

Individuals have a finite but unknown lifespan. The efficient allocation of financial resources across an unknown lifespan is a planning challenge because consumption smoothing requires the allocation of available financial resources across an expected time frame. Humans may plan to spread their resources based on an average lifetime, but this strategy exposes them to the risk of outliving their assets in old age. One efficient strategy is to pool the risk of an unknown lifespan across individuals through the use of an annuity.

Annuities have existed in a variety of forms for thousands of years. The Romans sold a financial instrument called an “annua” that returned a fixed yearly payment, either for life or for a specified period, in return for a lump sum payment. Even today, annuities remain popular risk management tools, especially for older individuals and retirees who want to mitigate the risk associated with outliving their assets (i.e., longevity risk). Annuities are generally purchased from an insurance company; however, government pensions and payouts from employer pension plans are also technically annuities.

Annuities have become increasingly complex over time, and various types exist. With an **immediate annuity**, an amount of money is paid to the insurance company in return for a guarantee of specified future monthly payments over a specified period of time, either a number of years or the life of the insured, who is called the *annuitant*. A **deferred annuity** allows an individual to purchase an income stream to begin at a later date. We further discuss various forms of immediate and deferred annuities in the following sections.

The annuity payment guaranteed by the insurance company is most directly based on the amount of money tendered, the age and gender of the annuitant, and the insurance company’s required rate of return (including its cost of funds and its expense and profit factors). Although life insurance helps provide financial protection in case the insured “dies too soon,” immediate life annuities provide financial protection in case the insured “lives too long.”

In this section, we discuss some of the key aspects of annuities, such as the parties to an annuity contract, the different types of annuities that exist, payout methods, annuity taxation, and the appropriateness of annuities for individuals. We conclude with an overview of how annuities can benefit retirees and provide a framework that explains which retirees may benefit the most from an annuity.

Parties to an Annuity Contract

There are four primary parties to an annuity contract: the insurer, the annuitant, the contract owner, and the beneficiary. The insurer—generally an insurance company—is the entity that is licensed to sell the annuity. The annuitant is the person who receives the benefits. The contract owner is the individual who purchases the annuity and is typically the annuitant. In some instances, the contract owner and the annuitant may be different: For example, if the annuity is purchased by a company for a retiring employee, then the company is the contract owner and the employee is the annuitant. Lastly, the beneficiary is an individual or entity that will receive any proceeds upon the death of the annuitant. For contracts like a plain vanilla single-premium annuity, there may not be any death benefit; however, with variable annuities and annuities

with some kind of minimum guaranteed payment period (e.g., 10 years), often referred to as a “period certain” option, there may be some residual value once the annuitant passes away.

Classification of Annuities

There are a variety of ways to classify annuities. The two most critical dimensions that help distinguish the primary types of annuities are (1) deferred versus immediate and (2) fixed versus variable. We will expand on these types shortly, but briefly, deferred annuities provide income that begins at a future date after the initial purchase of the annuity. In some cases, the original investment may retain some liquidity prior to initiation of annuity payments if the purchaser retains the right to sell the deferred annuity. In contrast, with immediate payout annuities—or single-premium immediate annuities (SPIAs), as they are often called—the individual permanently exchanges a lump sum for a contract that promises to pay the annuitant an income for life. For both deferred and immediate annuities, the annuity can be invested in what is termed a “fixed” account or a “variable” account.

Many other versions of annuities are available to meet different needs. For example, a *joint life annuity* is based on the expected lifespans of two annuitants, usually husband and wife, and payments are made as long as at least one of the two is alive. All else equal, the monthly payment is lower for this type of annuity because adding a second annuitant extends the payment timespan.

Deferred Variable Annuities

In its most basic form, a deferred variable annuity is similar to a mutual fund, although it is structured as an insurance contract and typically sold by someone licensed to sell insurance products. With most deferred variable annuities, there is a menu of potential investment options from which an individual can choose. Typical investment options might include a pre-determined target risk asset allocation consisting of a diversified mix of securities managed by multiple investment managers. Many of the investment managers replicate popular mutual fund strategies for the annuity separate account. Compared with traditional investment programs (e.g., mutual funds), these annuities can be more expensive for investors, and the number of investment fund options within the programs may be limited.

Deferred variable annuity contracts may include a death benefit. A typical death benefit guarantees that the beneficiary named in the contract will receive the entire amount used to purchase the annuity—a feature that has value to the beneficiary only if the individual dies when the value of the contract is less than the initial investment. Like all features of deferred annuities, the death benefit creates a risk for the issuing insurance company. To offset this risk, the insurance company charges a fee. As with mutual funds, individuals maintain control of their money through the right to exit (or sell) the contract, although there can be considerable surrender charges for withdrawing one’s money. Also, similar to mutual funds, a deferred variable annuity does not guarantee lifetime income unless the individual (1) adds an additional feature (a contract rider) or (2) annuitizes the contract by converting the value of the deferred variable annuity into an immediate payout annuity. It is worth noting that relatively few deferred variable annuity investors end up “annuitizing.”

Adding a guaranteed minimum withdrawal benefit for life rider to a deferred variable annuity can create a guaranteed income stream for life for the investor. The typical guaranteed minimum withdrawal benefit for life promises to pay the individual a fixed percentage (e.g., 4%) of the initial investment value as long as he or she lives. Each payment is subtracted from the current value of the deferred variable annuity contract. If the markets continue to perform well, the initial investment value may not be depleted, and any remaining value will go to the investor’s beneficiaries. In a

down market, the investment value may be depleted. If this is the case, the insurance company is contractually obligated to continue to pay the investor the guaranteed minimum benefit as long as the investor is alive.

Deferred Fixed Annuities

Deferred fixed annuities provide an annuity payout that begins at some future date. For each dollar invested, the insurance company will tell the investor how much income he or she will receive when annuity payments commence at a specified age in the future. It costs considerably less for a 30-year-old to purchase a dollar of income for life starting at age 65 than it does for a 55-year-old to purchase a dollar of income for life starting at age 65. At any point prior to annuitization (i.e., conversion of the investment into an annuity), the investor can cash out and receive the economic value of the accumulated purchases less any applicable surrender charges, in which case the annuity contract is terminated. Once in retirement, the individual has two options: (1) cash out or (2) begin withdrawing the accumulated funds. In either case, the “economic value” of the accumulated purchases is annuitized, converting the deferred fixed annuity into an immediate fixed. In contrast to deferred variable annuities, which most investors choose not to annuitize, most deferred fixed annuities are eventually annuitized.

Immediate Variable Annuities

With an immediate variable annuity, the individual permanently exchanges a lump sum for an annuity contract that promises to pay the annuitant an income for life. As the name suggests, the amount of the payments varies over time based on the performance of the portfolios that the assets are invested in. A common feature that can be added to an immediate variable annuity for an additional cost is an income floor that protects the annuitant in the event of a down market.

Immediate Fixed Annuities

With immediate fixed annuities, the most common and most utilized type of annuity, an individual trades a sum of money today for a promised income benefit for as long as he or she is alive. The “income yield” for an immediate fixed annuity is the total amount of ongoing annual income received as a percentage of the initial purchase price. For example, if an individual purchases an immediate fixed annuity for \$100,000 and in exchange receives a guarantee to be paid \$8,000 per year for as long as the individual is alive, the income yield for the annuity would be 8%.

The income yield for immediate fixed annuities, or any type of annuity, varies based on a number of factors. One key factor is the age of the insured individual (or individuals). Exhibit 7 contains the payout rates for two different immediate annuity types: a life-only annuity (which pays benefits only as long as the individual is alive, with no residual benefits) and a life annuity with a 10-year certain payment (whereby benefits are guaranteed to last for at least 10 years). Quotes are included for three different types of annuitants: male, female, and joint (a couple). (The couple consists of a male and a female assumed to be the same age, and the survivor benefit is 100% of the primary benefit; that is, the benefit stays the same as long as either of the couple survives.)

Exhibit 7: An Example of Annual Payouts as a Percentage of Initial Premium

Life Only				Life with 10-Year Period Certain			
Age	Male	Female	Joint	Age	Male	Female	Joint
60	6.28%	5.87%	5.51%	60	6.15%	5.86%	5.42%
65	7.02	6.47	5.96	65	6.75	6.32	5.88
70	8.04	7.31	6.65	70	7.46	7.01	6.59
75	9.53	8.73	7.68	75	8.33	7.93	7.45
80	11.90	10.87	9.35	80	9.30	8.96	8.51
85	15.17	14.27	11.70	85	10.08	9.95	9.45
90	20.10	19.34	14.51	90	10.66	10.49	9.86

Source: www.immediateannuities.com (retrieved December 2014).

There are a number of important takeaways from Exhibit 7. First, the payouts (i.e., income yields) are higher when expected remaining longevity is shorter. For example, a male of 85 will receive a higher income yield than a male of 65 because the older male has a shorter life expectancy. A 65-year-old female will have a smaller payout than a 65-year-old male because females have a longer average life expectancy than males. The income yield is determined by estimating the average longevity of a given annuitant pool. A shorter average payment period will mean higher income for the older annuitants in the pool.

Another takeaway from the table is that the inclusion of a period certain (or return-of-premium feature) will reduce the payout, but to varying degrees. For example, adding the 10-year period certain has a relatively small effect on the income yield for a 60-year-old male (which decreases from 6.28% to 6.15%), but it has a significantly greater impact for the 90-year-old male (whose payout rate decreases from 20.10% to 10.66%). Again, this difference is based on life expectancies. The probability of a 60-year-old male dying in the first 10 years is much smaller than the probability of a 90-year-old male dying over the next 10 years, so the payouts are adjusted accordingly. An individual who is concerned with lifetime income maximization is likely better off not adding any type of rider that includes a residual benefit; however, such a rider may be desirable if an individual has competing goals of generating lifetime income and providing some residual wealth for heirs.

In addition to mortality, another key variable that affects annuity pricing is the expected return the insurance company can earn on premiums. Because insurance companies tend to invest conservatively, the available yield on bonds provides a relatively good proxy, at least from a historical perspective, for how payout rates change over time. When current yields on bonds are lower than historical bond yields and life expectancies are increasing, payouts on annuities will be relatively low by historical standards. Low annuity yields may discourage many individuals from buying annuities if they believe that yields will eventually go back up. In reality, however, lower annuity payout rates simply reflect the increasing cost of hedging longevity risk using available investments.

Advanced Life Deferred Annuities

The final type of annuity that we discuss is a hybrid of a deferred fixed annuity and an immediate fixed annuity. This so-called *advanced life deferred annuity* (ALDA) is often referred to as pure longevity insurance. Although it might sound a bit contradictory, ALDAs are deferred immediate payout annuities. Similar to an immediate fixed annuity, an ALDA involves the permanent exchange of a lump sum for an insurance

contract that promises to pay an income. However, in contrast to an immediate payout annuity, for which the payments begin immediately, an ALDA's payments begin later in life—for example, when the individual turns 80 or 85.

Given that a specific monthly benefit may not begin until age 85, a deferred immediate life annuity would clearly cost less to purchase than a regular immediate life annuity. We note three reasons for this lower cost. First, because payments on the deferred annuity begin so far in the future, the insurance company has ample time to earn money on the amount tendered. Second, life expectancy for an 85-year-old is much shorter than for a 65-year-old, so the number of payments made will be fewer. Third, it is quite possible that the annuitant may actually die before any payments are made. For a relatively small premium, longevity insurance can provide additional security and can supplement income in later years.

EXAMPLE 12

Comparing Annuities

1. Compare fixed immediate annuities and variable immediate annuities.

Solution:

Both fixed and variable immediate annuities represent an irrevocable exchange of money for an insurance contract (the annuity contract). With a fixed immediate annuity, payments are “fixed” in either nominal terms or, in some cases, real terms, providing certainty about payment streams. With both fixed and variable immediate annuities, a common feature is a “period certain,” whereby the payments continue to a designated beneficiary for a specified period, typically 10 years.

10

ADVANTAGES AND DISADVANTAGES OF FIXED AND VARIABLE ANNUITIES

- ☐ discuss the use of annuities in personal financial planning
- ☐ discuss the relative advantages and disadvantages of fixed and variable annuities

In this section, we discuss the relative advantages and disadvantages of both fixed and variable annuities. As a reminder, fixed annuities provide a benefit that is fixed (or known) for life, whereas variable annuities have a benefit that can change over time and that benefit is generally based on the performance of some underlying portfolio or investment. When selecting between fixed and variable annuities, there are a number of important considerations.

Volatility of Benefit Amount

The most obvious difference between fixed and variable annuities is the type of benefit. Fixed annuities provide a constant income stream that is guaranteed not to change, whereas the income from a variable annuity could change considerably depending

on the terms of the annuity payout. Retirees seeking a high level of assurance with respect to benefit payouts are likely better served by a fixed annuity, or a variable annuity that limits the possible change in the benefit over time. Retirees who are risk tolerant may be more interested in a variable annuity. If a retiree is willing to adjust his or her spending over time, that individual may be able to increase the amount spent each year by selecting a variable annuity for which the payment is linked to a risky portfolio of assets.

Flexibility

The flexibility of an annuity varies materially with the type of annuity and its individual features. For example, an individual who purchases an immediate fixed annuity has effectively traded some amount of wealth for a guarantee of income for life. In most situations, this exchange is irrevocable: The individual who purchased the immediate fixed annuity cannot “undo” the transaction and request the original purchase amount back. The fact that these annuities are irrevocable makes sense from the insurer’s perspective because (in theory) if given the option, every individual would request the initial premium back.

Variable annuity payments are typically tied to the performance of an underlying subaccount. This subaccount can often be withdrawn by the annuitant, subject to limitations. Therefore, variable annuities can provide the annuitant with guaranteed income for life as well as the flexibility to access the funds should he or she (or they) need to do so. There may be penalties associated with withdrawing funds from a variable annuity, and in some cases, withdrawals may not be allowed (e.g., in the case of an immediate variable annuity).

Future Market Expectations

A fixed annuity locks the annuitant into a portfolio of bond-like assets at whatever rate of return exists at the time of purchase. This scenario creates some interest rate risk because the value of these underlying securities will fall if interest rates rise.

If the annuitant assumes that interest rates will vary over time, he or she may be tempted to delay annuitization until interest rates rise. This delay, however, will reduce expected consumption during the delay period because the annuitant receives only market returns on investment assets and no mortality credits. Mortality credits, which will be discussed at greater length shortly, are effectively the difference between the future payout one would receive without pooling one’s investments and the future payout one receives when the investment pool includes individuals who have already passed away. Delay of annuitization may also expose the annuitant to the risk that life expectancies will systematically increase during the delay period, which would increase the cost of annuitization (i.e., decrease future potential available payouts). Annuitizing earlier allows the annuitant to hedge the risk of a large increase in future longevity for a population.

Variable annuities allow an annuitant to accept some variation in payments in return for the possibility of higher future payments if the market performs well. Because most retirees will rationally accept some market risk with their investment portfolio, this option allows individuals to increase their retirement income efficiency by benefiting from both a mortality credit and a risk premium. The benefit of accepting market risk varies by type of annuity, and many variable annuity features limit the potential growth in income payments in future time periods. Variable annuities without growth-limiting features, such as minimum income guarantees, are most likely to provide a future income that outpaces inflation on average.

Fees

The fees associated with variable annuities tend to be higher than those for fixed annuities. These higher fees come from a variety of factors but are primarily attributable to the costs of hedging market risk, administrative expenses, and reduced price competition. Evidence suggests that the price of insurance products is significantly affected by price competition, and immediate fixed annuities are much easier for a consumer to compare (the consumer will simply select the highest annuity payment for the amount he or she will spend). The opaque pricing of variable annuities can reduce price competition if consumers are unable to easily compare the relative efficiency of product characteristics. A thorough price analysis may require weighing the possible added costs of opaque variable annuity features—perhaps using analytical tools to help assess these costs—against the benefits of a potentially higher return.

Inflation Concerns

Inflation can have a significant negative impact on the real income received from a fixed annuity. For example, if annual inflation averages 3%, after approximately 24 years, the income would be worth approximately half as much as it was worth when the annuity began. In their purest form, fixed annuities are nominal and will not change with inflation; however, it is possible to create a partial inflation hedge by having benefits “step up” some predetermined amount each year (e.g., 3%). Although this adjustment is not a perfect inflation hedge, it does provide a mechanism to ensure that payments increase over time. There are a number of variable annuities (and riders on fixed annuities) that allow the payments to increase or decrease based on changes in inflation. An individual interested in guaranteeing some lifetime level of income that changes with inflation may find these types of policies or riders valuable.

Payout Methods

The payout methods available from an annuity are similar regardless of whether the annuity is fixed or variable. In certain instances, the annuitant is unable to choose the payout method, especially with some types of government pensions, but there is generally some level of choice when purchasing a private annuity from an insurance company. The primary payout methods were discussed previously but are summarized again here:

- *Life annuity*: Payments are made for the entire life of the annuitant and cease at his or her death.
- *Period-certain annuity*: Payments are made for a specified number of periods without regard to the lifespan or expected lifespan of the annuitant.
- *Life annuity with period certain*: This payment type combines the features associated with a life annuity and a period-certain annuity, so payments are made for the entire life of the annuitant but are guaranteed for a minimum number of years even if the annuitant dies. The most common length of the period certain is 10 years. For example, if the annuitant dies after 6 years, a life annuity with 10 years period certain will make payments to the annuitant's beneficiary for the remaining 4 years and then cease.
- *Life annuity with refund*: This type is similar to a life annuity with period certain, but instead of guaranteeing payments for life or for a certain number of years, a life annuity with refund guarantees that the annuitant (or the beneficiary) will receive payments equal to the total amount paid into the contract, which is equal to the initial investment amount less fees.

- *Joint life annuity:* With a life annuity on two or more individuals, such as a husband and a wife, payments continue until both members are no longer living. For example, a married couple may purchase an immediate annuity that pays a monthly benefit as long as either one of them is alive. The contract states the benefit that the survivor will receive, which can be as much as 100% of the primary benefit or a smaller amount, such as 50% or 75% of the full benefit. The annuity payments cease when the survivor passes away.

It is important to note that the payout methods are not mutually exclusive. In theory, one could combine each of the different methods into a single annuity. Annuity payments can also be made at different frequencies, such as monthly, quarterly, or annually, although monthly payments are generally the most common. It is also possible to include riders or other methods that specify how the benefit may change over time. For example, it is possible to purchase an annuity for which the payment increases by some fixed percentage (or amount) each year.

Annuity Benefit Taxation

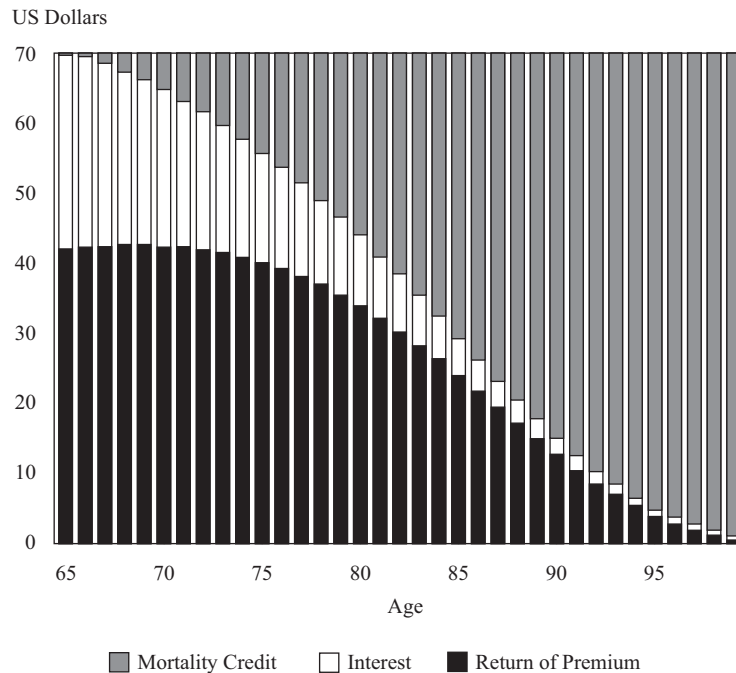
In some locations, annuities can offer attractive tax benefits, such as tax-deferred growth. For example, in the United States, the growth in an annuity is taxed only when the individual receives income from the annuity. This presents an opportunity for tax-deferred growth, especially for someone who purchases the annuity at a relatively young age or has exhausted other tax-sheltering alternatives. The actual taxation of the benefits varies considerably by type of annuity but is generally based on some average of the difference between the amount paid for the annuity and the benefits received.

The actual method of taxation varies materially by country, and before purchasing an annuity or recommending an annuity for a client, one should become familiar with the applicable tax consequences or consult a local expert on annuity taxation. In general, the potential for tax deferral, combined with a high marginal tax rate on alternative investments, may make annuities attractive for retirees.

Appropriateness of Annuities

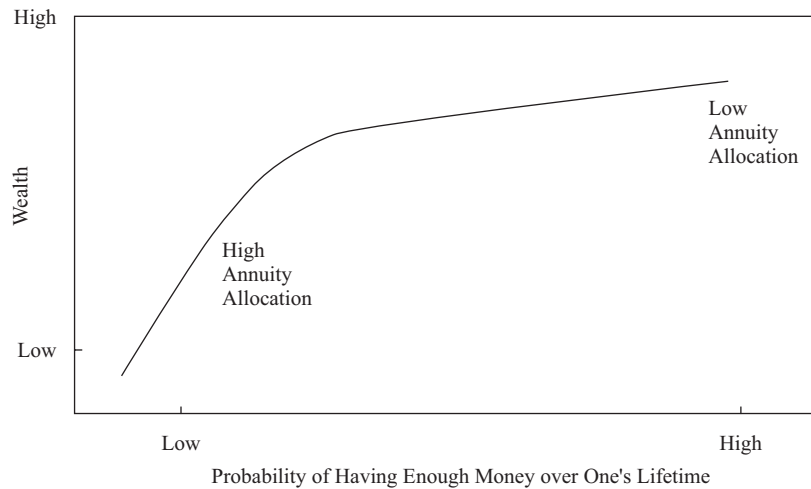
When creating an income stream from a pool of assets, each retiree has a choice. The individual can choose either to receive periodic withdrawals from an investment portfolio (i.e., not annuitize) or to purchase an annuity (i.e., annuitize). This decision is obviously based on a variety of factors and preferences. When discussing the potential benefits of annuitizing, it is important to understand the concept of mortality credits (which we briefly mentioned earlier). Each payment received by the annuitant is a combination of principal, interest, and mortality credits. Mortality credits are the benefits that survivors receive from those individuals in the mortality pool who have already passed away. It is possible to demonstrate this effect visually using a chart (see Exhibit 8).

In Exhibit 8, we show an example of mortality credits for US male individuals, using male mortality rates based on the Society of Actuaries' 2012 Individual Annuity Reserve Table and an interest rate of 3%. We also assume that the individual receives US\$70 of annual income for life for an approximate initial cost of US\$1,000. Given these assumptions, the annual benefit payment can be decomposed into three parts: interest (based on the remaining assets from the initial investment minus benefits paid), return of premium, and mortality credits.

Exhibit 8: Mortality Credits

An individual who self-insures longevity risk would receive only (approximately) the interest and return-of-premium portions of Exhibit 8. The additional mortality credits arise because some individuals will pass away early, thereby subsidizing the future benefits of those individuals who are still alive. In this way, an individual can receive more income and additional certainty from purchasing an annuity. However, this certainty comes at a cost because the expected benefits of an annuity are generally not positive. Annuities are a form of insurance, and most forms of insurance, by definition, do not (and should not) have a positive expected value because it would imply that the insurance company lost money on the average policy sold. Moreover, it would likely not be advantageous to purchase an annuity from an insurance company that lost money on the average policy sold because that company would likely eventually go out of business and no longer be able to fund future expected benefit payments.

Therefore, an individual who purchases an annuity is acquiring the benefit of certainty regarding lifetime income in exchange for accepting lower potential wealth at death, as well as lower lifetime income depending on the cost of the annuity. This perspective can be used to create a “retirement income efficient frontier” whereby the decision of how much to annuitize is based on an individual’s preference for wealth maximization and aversion to running out of money. This concept is similar to the traditional Markowitz (1952) efficient frontier; but for the retirement income efficient frontier, wealth replaces expected returns on the vertical axis and shortfall risk (which is defined approximately as the risk associated with running out of money over one’s lifetime) replaces standard deviation on the horizontal axis. The exact definitions of “wealth” and “shortfall risk” (as well as the terms themselves) have varied across literature but still tend to be relatively consistent. The concept of the retirement income efficient frontier, which is displayed visually in Exhibit 9, is derived from Chen and Milevsky (2003), among others.

Exhibit 9: Retirement Income Efficient Frontier

Certain factors can generally be expected to affect a retiree's demand for annuities, either positively or negatively. For example, the following factors would generally suggest increased demand for an annuity:

- Longer-than-average life expectancy
- Greater preference for lifetime income
- Less concern for leaving money to heirs
- More conservative investing preferences (i.e., greater risk aversion)
- Lower guaranteed income from other sources (such as pensions)

The broad international shift away from defined benefit plans and toward defined contribution plans has increased the demand for annuities. Therefore, it is important to have an understanding of annuities and which individuals they are most likely to benefit.

RISK MANAGEMENT IMPLEMENTATION

11

A variety of factors need to be considered when implementing an optimal risk management strategy for an individual or a household. In this section, we will indicate how human capital affects the asset allocation decision, provide a case study analyzing and critiquing an insurance program, and present appropriate strategies for asset allocation and risk reduction for a given investor.

Determining the Optimal Risk Management Strategy

As with portfolio selection, the decision to retain risk or to manage risk through insurance or annuities is determined by a household's risk tolerance. At the same level of wealth, a more risk-tolerant household will prefer to retain more risk—either through higher insurance deductibles or by simply buying little or no insurance—than a less risk-tolerant household. Additionally, the amount of overhead and administrative expenses built into the cost of the insurance (the load) will vary by product.

If an individual decides not to insure a risk exposure, he or she may still choose to moderate the impact of a potential loss. The term *loss control* refers to efforts to reduce or eliminate the costs associated with risks. There are three general approaches to loss control. *Risk avoidance* is the purest form of loss control. That is, one can remove the possibility that an event involving loss will occur. For example, one may avoid the risk of loss of a collector car or piece of jewelry by selling the asset. This strategy can be particularly appealing if the asset is no longer providing significant utility, or if the magnitude of the risk exposure rises because of price appreciation. Two other types of loss control are *loss prevention* and *loss reduction*. Loss prevention is the process of taking actions to reduce the probability that a loss event will occur. For example, installing a security system reduces the probability of a break-in that could lead to a loss by theft. Operating an automobile that has a backup camera reduces the probability of causing property damage or personal injury when the automobile is backing up. Installing a swimming pool alarm reduces the probability that a child will drown. Loss reduction is the process of seeking to reduce the size of a loss if a loss event occurs. For example, maintaining a high-quality fire extinguisher in the kitchen may allow a homeowner to contain a fire that would otherwise cause extreme damage. Unlike the loss prevention examples, the fire extinguisher does not reduce the likelihood of the loss event.

As initially discussed in Section 3.1.3, in addition to risk avoidance and risk reduction, individuals can also manage risk through the techniques of *risk transfer* and *risk retention*. Although risk transfer generally involves insurance or annuities, individuals can also use *non-insurance risk transfers* in many situations, and these usually take the form of contracts. For example, an apartment renter may select a long-term lease to lock in the amount of the monthly rent for a longer time during a period of inflation, thus transferring the risk of increased rent to the landlord, who then must absorb the opportunity cost. Incorporation of an individual's business provides a non-insurance risk transfer in many countries. By incorporating, the individual shields his or her personal financial assets from any legal judgment in which the business is found to be at fault.

A systematic risk management approach would be to consider the optimal strategy for each risk exposure. Some guidelines are helpful in deciding when it is appropriate to accept (retain) risk, when it is best to reduce the potential magnitude (or severity) of the risk, and when it makes sense to transfer risk. Small-magnitude risks below a risk retention limit should be retained (i.e., self-insured) and the magnitude/severity reduced to the point where the expected cost is equal to the expected benefit. For example, installing a high-quality fire-rated roof on a house may reduce both the probability of a fire (e.g., from sparks from a house fire next door) and the potential damage from a fire. However, the cost of installing a new, expensive roof may be too high to be justified. In this case, the large-magnitude potential loss from fire can be transferred through property insurance and a deductible can be used to set the optimal amount of risk retention.

A common risk management approach is summarized in Exhibit 10.

Exhibit 10: Risk Management Techniques

Loss characteristics	High frequency	Low frequency
High severity	Risk avoidance	Risk transfer
Low severity	Risk reduction	Risk retention

An insurance product can provide indemnification by ensuring that there will be no (or minimal) loss in economic net worth. For example, an insurance claim will provide reimbursement of flood damage to a \$20,000 vehicle if the household has a comprehensive insurance policy. If not, the household risks reducing its economic net worth by the size of the loss exposure. Indemnification in life insurance can be more complex. A human life value analysis seeks to replace the loss in human capital. A needs analysis estimates the present value of future consumption that would need to be replaced if the income of a primary earner were lost.

EXAMPLE 13

Risk Management Strategy

1. Describe *loss control* in risk management.

Solution:

Loss control refers to efforts to reduce or eliminate costs associated with risks. A simple method of loss control is to avoid risks (risk avoidance)—for example, by not engaging in high-risk hobbies, such as rock climbing. Another approach to loss control is loss prevention, in which one attempts to reduce the likelihood of a risky event. For example, one may install an alarm system to discourage burglars. Loss reduction refers to approaches that attempt to minimize the size of the loss if a risky event does occur. For example, an airbag in an automobile does not reduce the likelihood of an accident, but it may reduce the seriousness of injuries sustained by those involved in an accident.

Analyzing an Insurance Program

This section provides a case study of how one might analyze an individual's insurance needs and design an insurance program. In this case, Jacques and Marion Perrier are 40 years old and 38 years old, respectively. They have two children: Henri, age 8, and Émilie, age 6. Jacques is a manager of technical services for a large corporation and earns €100,000 per year. Marion works part-time as a nurse, earning €20,000 after tax per year but plans to return to full-time work in 10 years, when Émilie turns 16. Marion expects that, with adjusted market conditions and after 10 years of inflation, she will earn €60,000 after tax per year as a full-time nurse at that time. Jacques and Marion are in excellent health and maintain a lifestyle that is well within their income.

The family lives in the city in a comfortable condominium that they bought for €250,000 five years ago and that is currently valued at €300,000. They owe €190,000 on a mortgage that still has 25 years to maturity. Although both Jacques and Marion take public transportation to work, they have an automobile, which is 10 years old but in excellent condition with relatively few kilometers of use. Jacques and Marion intend to live in the condominium at least until Jacques's planned retirement at age 60. At that time, they will decide whether to remain in the condominium or move to the small town nearby where Jacques grew up.

Four years ago, when his parents died, Jacques inherited the moderately sized but attractive home where he grew up. It was worth €150,000 at the time and has increased in value to €165,000 (a 10% increase during this period). For two years, the family tried to use the house as a weekend retreat, but they discovered that their children's activities and the desire to go to different places caused them to use the house infrequently. For the past two years, they have rented the house to a middle-aged couple

who have no children. There is no debt on the house, and the rent is enough to pay for taxes and other expenses and to generate positive cash flow that Jacques and Marion are using for family vacations.

Both of Jacques's parents died at age 70. Marion's father died at age 80. Her mother, Françoise, is in good health at age 72, and women in her family have generally lived to very old ages. Françoise has a pension but does not have much in assets.

Current Insurance Plan

Life Insurance.

Jacques bought a whole life insurance policy with a death benefit of €200,000 when Henri was born. Jacques is the insured, and his estate is the beneficiary. His employer also provides a €50,000 term life insurance policy that names Marion as the beneficiary. There is no life insurance on Marion.

Health Insurance.

The family is covered by a national health insurance plan.

Disability Insurance.

Jacques's employer provides short-term disability payments for up to six months. As a part-time employee, Marion has no disability benefits.

Long-Term Care Insurance.

Elderly individuals in Jacques and Marion's country are eligible for long-term care at a cost equal to 75% of their pension benefits.

Property Insurance.

The exterior of the condominium is insured through the condominium owners' association. The insured value is increased annually to consider current market and replacement values. Jacques and Marion have a property insurance policy on the contents of their condominium for €20,000. Jacques's parents insured the house for €100,000 15 years ago. They never increased the coverage, and Jacques has maintained the same policy. The automobile has coverage for liability and collision, as well as comprehensive coverage.

Data Summary

- Jacques: age 40; €100,000 annual earnings; €200,000 whole life insurance policy; €50,000 term life insurance policy
- Marion: age 38; €20,000 annual earnings after tax; no life insurance
- Children: Henri, age 8, and Émilie, age 6
- Condominium: €300,000 current value; €190,000 25-year remaining mortgage; exterior of building fully insured; contents insured for €20,000
- Rental home: €165,000 current value; no mortgage; insured for €100,000

Program Review

Life Insurance.

As we discussed earlier, two approaches are commonly used in calculating life insurance needs. The first is the human life value method, which estimates the present value of earnings that must be replaced. As we will demonstrate, the human life value method indicates that Jacques should purchase approximately €1.11 million of additional life insurance. The second approach is the needs analysis method, which

estimates the financial needs of dependents. As we will also demonstrate, the needs analysis method indicates that Jacques should purchase approximately €1.25 million of additional life insurance.

Of course, there is no absolute when calculating life insurance needs because there are so many variables and uncertainties. However, in this case, the two methods generate relatively close numbers. To augment Jacques's current life insurance program, he should purchase at least €1 million in additional term life insurance, but €1.25 million would be more desirable.

Although an annually renewable term policy will be less expensive, a 20-year level term policy should be considered. At the end of 20 years, Jacques will be nearing the end of his working career and the children will be fully grown. Jacques and Marion will have accumulated a high enough level of savings that income replacement will be unnecessary. At that time, the term policy can be allowed to lapse and the €200,000 whole life policy can be retained.

Whole life coverage like Jacques's current policy is relatively expensive, but the pricing and cash build-up at this point in the policy's life indicate that it is probably wise not to replace the policy with term coverage now. Even though life insurance rates have fallen as life expectancies have lengthened and competition has increased, the decline is probably not enough to offset the advantages of the whole life policy at this point. In addition to providing life insurance coverage, the policy's cash-value buildup can be considered a conservative part of the asset allocation of the retirement portfolio.

Marion should be named as the primary beneficiary on the new policy, with Henri and Émilie as secondary beneficiaries, although countries may differ on the exact approach for designating a beneficiary who is a minor. It may be advisable to name a custodian and/or create a trust for Henri and Émilie until they reach adulthood. The old policy should be changed to include these beneficiary designations as well. Making death benefits payable to the estate, as is the case currently on the whole life policy, can make receipt of payments much slower and may subject benefits to greater taxation.

Although Marion's income is modest, her death would cause a financial burden in terms of both lost income and added expenses to replace her family and household responsibilities. Furthermore, her income will increase when she returns to full-time work in 10 years, and financial responsibility for the children will still exist at that time. At Marion's current age and with her excellent health, and because she is female, life insurance rates are very low. However, it is possible that a health issue or accident could cause her rate to increase or even make her uninsurable. Given the complexity of Marion's situation, it is difficult to calculate a recommendation for the amount of life insurance needed.

Life insurance companies typically offer lower rates for larger amounts of coverage in a series of steps. For example, the price per €1,000 of coverage may decrease at €100,000, €250,000, €500,000, and €1,000,000 of coverage. But companies are hesitant to underwrite policies that are large relative to the insured's human life value, or the discounted value of future earnings. A €250,000 policy would take advantage of the price break but would be relatively large for Marion's current income. Still, it is justifiable based on her future employment plan. Purchasing a policy of this size will provide adequate protection for the 10 years until she goes back to work full time, and at that point the children will be more independent, reducing the need for life insurance. As with Jacques, a 20-year level term policy is reasonable.

LIFE INSURANCE NEEDS FOR JACQUES AND MARION

As mentioned previously, the *human life value* method is consistent with the concept of human capital and involves replacing the estimated net contributions to family finances that Jacques would generate if he did not die during his projected earning life. Calculating these contributions involves the following steps:

- Start with the actual pre-tax compensation that Jacques would receive from employment: €100,000.
- Adjust for income taxation, and here we assume a 30% rate: €100,000 – €30,000 = €70,000 post-tax compensation.
- Adjust for family expenses attributable to Jacques that will not exist after his death, such as his transportation, travel, clothing, food, entertainment, and insurance premiums. Here, we assume those expenses to be €20,000. So €70,000 – €20,000 = €50,000 income after expenses.
- Add the value of any non-taxable employee benefits that the family will no longer receive, such as employer contributions to retirement plans, which we assume to be €15,000: €50,000 + €15,000 = €65,000.
- Estimate the amount of pre-tax income needed to replace that income on an after-tax basis. Note that the rate of taxation of annual income generated from life insurance proceeds may be different from the rate of taxation of Jacques's employment income, and marginal rates may be lower for lower incomes. Here, we assume a 20% tax rate: $€65,000 / (1 - t) = €65,000 / (1 - 0.20) = €81,250$.
- We then apply an annual growth rate, assumed here to be 3%, to consider the effects of inflation and career advancement over the full 20 years until retirement.
- Finally, we discount all the future cash flows back to the present at an appropriate rate, assumed here to be 5%.

Assuming the lost income replacement would be needed by Jacques's family immediately, the human life value calculation can be solved as the present value of an annuity due with growing payments (a so-called "growing annuity due"). Using calculator keystrokes for an annuity due with level payments, the growth of payments can be incorporated by adjusting the discount rate to account for the growth rate of earnings. The adjusted rate i can be calculated as follows:³ $[(1 + \text{Discount rate}) / (1 + \text{Growth rate})] - 1$, or $(1.05/1.03) - 1 = 1.94\%$. Thus,

- set the calculator for beginning-of-period payments;
- $n = 20$ (the number of years until retirement);
- payment = €81,250; and
- $i = 1.94\%$.

Solving for the present value of an annuity due, the human life value method recommends €1,362,203 of life insurance for Jacques. Because Jacques already has €250,000 of life insurance, he should purchase an additional €1,112,203, according to this method. This amount would likely be rounded to €1.1 million.

³ Most individuals' situations would call for a discount rate that is higher than the inflation rate. If that is not the case, or if future cash flows are not projected to grow at a constant rate, the present value can be calculated in the typical manner as the present value of a series of unequal payments.

As discussed earlier, the *needs analysis* method is concerned with meeting the financial needs of the family rather than replacing human capital. Needs analysis typically includes the following steps, which are presented in greater detail in Exhibit 11:

- Estimate the amount of cash that will be needed upon the death of the insured person. This amount will include final expenses (funeral and burial) as well as any taxes that may be payable. It is also common to pay off all debt (including mortgages) and to fully fund future education costs. An emergency fund should be created.
- Estimate the capital needed to fund family living expenses. This calculation requires discounting estimated living expenses (i.e., calculating the present value of future cash flow needs) during multiple time frames, typically as follows:
 - Estimate the surviving spouse's living expense needs, assumed here to continue for 52 years, until Marion is 90 years old. Note that when the mortgage and other debts are paid off, living expenses are lower.
 - Estimate the children's living expense needs, assumed here to continue until they are 22 years old. This amount does not include the education fund.
 - Include an additional amount for extra expenses during a transition period after Jacques's death, perhaps covering two years. In general, this period recognizes that there may be some contractual obligations, such as a car lease, that may not terminate upon a person's death.
 - Consider Marion's future income (earnings). Note, however, that Marion may prefer not to go back to work full time as soon as planned because of the extra responsibilities of being a single parent. She may even choose to resign from her part-time job.
- Calculate total needs as the sum of cash needs and capital needs.
- Calculate total capital available, which may include cash/savings, retirement benefits, life insurance, rental property, and other assets.
- Calculate the life insurance need as the difference between the total financial needs and the total capital available.

Exhibit 11 is a representation of a needs analysis for Jacques.

Exhibit 11: Financial Needs: Life Insurance Worksheet

Cash Needs	Euro (€)
Final expenses	10,000
Taxes payable	5,000
Mortgage retirement	190,000
Other debt	10,000
Education fund	200,000
Emergency fund	30,000
Total cash needs	445,000
Capital Needs [present value of annuity due: growth rate = 3%, discount rate = 5%, adjusted rate (as above) = 1.94%]	
Marion's living expenses (60,000/year for 52 years)	1,991,941
Children's living expenses:	
Henri (10,000/year for 14 years)	123,934
Émilie (10,000/year for 16 years)	139,071

Cash Needs	Euro (€)
Transition period needs (10,000/year for 2 years)	19,810
Less Marion's income:	
Until Émilie is 16 (20,000/year for 10 years)	–183,713
Age 48–60 (60,000/year for 12 years)	–398,565 ^a
Total capital needs	1,692,478
Total Financial Needs	2,137,478
Capital Available	
Cash and savings	30,000
Vested retirement accounts—present value	200,000
Life insurance	250,000
Rental property	165,000
Total capital available	645,000
Life insurance need (Total financial needs less total capital available)	1,492,478

^a *Calculated in two steps:* (1) Compute the amount needed in 10 years, when Marion will begin earning €60,000 per year. Assuming 12 years of earnings from age 48 to age 60, a 3% annual growth in earnings, and a 5% discount rate (1.94% adjusted discount rate), a present value of an annuity due calculation shows that €649,220 will be needed in 10 years. (2) Discount the €649,220 back to the present—10 years at the unadjusted discount rate of 5%—for a total of €398,565. The discount rate is not adjusted during this period because there are no payments to which a growth rate would be applied. We simply discount a future value to the present.

This amount of life insurance under the needs analysis method would likely be rounded to €1.5 million—considerably higher than the €1.11 million calculated earlier using the human life value method. The amount of life insurance selected may depend on which method seems to be more relevant to the family situation. One may view the two values as a reasonable range and use the larger number, the smaller number, or perhaps an average of the two. In many cases, selection of an amount may be further affected by “breakpoints” in the insurance company’s premium schedule. The premium rate may decline when insurance coverage reaches, for example, €250,000, €500,000, and €1,000,000.

As a final note, consider that if Jacques dies prematurely, there will be an increased need for life insurance for Marion while Henri and Émilie are still children.

Recommendations

Health Insurance.

Although the Perrier family is covered by national health insurance, they may want to seek private health insurance. In many countries, this type of coverage provides quicker treatment, a wider choice of physicians, and a higher standard of care.

Disability Insurance.

Both Jacques and Marion should consider long-term disability income insurance that guarantees the option to purchase additional coverage without underwriting. This type of policy would allow them to increase coverage as their incomes increase. They should look for “own occupation” coverage that specifies that they would be considered disabled if they could no longer perform the duties of their current positions. They should also select a benefit period that extends at least until their respective retirement ages, possibly age 65. For Jacques, a relatively long, 180-day elimination period would coordinate well with his company’s short-term disability plan and would

save money. Marion should consider a shorter elimination period, perhaps 90 days, because she has no short-term coverage. Both should choose an option that allows increased coverage based on inflation.

Taxation should be considered in purchasing a disability income policy. In some jurisdictions, premium payments may be deductible for income tax purposes but benefits are taxed. In others, it may make a difference whether the policy is purchased individually or through the employer. In any case, the amount of the benefit selected should be sufficient to replace income, net of any tax advantages and reduction of expenses. Most insurance companies do not like to sell policies with benefits that represent more than 70%–80% of an individual's income (less for high-income individuals) because high percentages create a moral hazard, in that the insured may decide that declaring a disability would give him or her a larger net income without the need to work.

Long-Term Care Insurance.

Although the Perriers' country provides some degree of long-term care assistance, it is possible that the facilities available are not of the standard that would satisfy them. Furthermore, the pricing structure appears to strongly favor individuals with limited income. The Perriers should consider long-term care insurance for themselves, especially if there is a coordination provision with the national plan. At their current ages, rates would be reasonable and would be locked in at the time the policy is purchased. Although stays in long-term care facilities are typically shorter than five years, it would be prudent to purchase a policy that does not have a time limit. Most policies have benefits that are based on a specified amount of money per day. The amount selected should be appropriate for the local cost structure, and the Perriers should pay the extra premium required to receive an automatic inflation adjustment, both before and after any claim.

Long-term care insurance may also be appropriate for Marion's mother, especially because women in her family tend to have long lives. This coverage will be more expensive at her age, but it would alleviate the potential need for care in the home, which would probably be provided by Marion. To avoid the financial and psychological stress of providing home care, a long-term care policy would be useful.

Property Insurance.

Jacques and Marion should check to make sure that the condominium association's insurance coverage is sufficient for the structure. They should also determine whether that policy provides any coverage for contents and, if so, to what extent. It is likely that the condominium association's coverage for contents, if any, is modest, and Jacques and Marion undoubtedly have personal property worth far more than the €20,000 of coverage in their personal policy.

The Perriers should make a thorough valuation of their personal property, even though this is a cumbersome task, and make sure they have or obtain a sufficient amount of coverage. A prudent approach that helps avoid insurance claim problems is to make a written and/or photographic inventory of all contents and improvements within the condominium. Any personal property that has specified limits within their policy should be appraised and scheduled. For example, an expensive necklace should be taken to a jeweler for a formal appraisal, after which the necklace should be added to the insurance policy, which will likely require a higher premium. A copy of the appraisal and inventory should be provided to the insurance company, and another copy should be kept off site in case of a disaster in their condominium.

Property insurance on the house should be reviewed. It appears that the amount of coverage is far less than appropriate. Furthermore, it is not clear whether the transfer of ownership of the property is properly noted on the policy. In the case of a claim, Jacques and Marion may have to deal with legal issues in order to collect. Even if

the property ownership issue is clarified, it is likely that the house is insured as the Perrier's residence or secondary residence. However, it is now a rental house, and it requires a different insurance policy. They should also determine whether the rental house's contents are included in the policy. Because they probably are not, it would be worthwhile to explain to the renters that they should obtain their own insurance on the contents (their property). This step is especially important because the Perriers could be considered liable for any loss of the renters' property if, for example, there is a fire caused by faulty wiring.

With regard to auto insurance, the Perriers should make sure that they have substantial liability insurance coverage, especially because it appears that they will have substantial assets to protect. However, they may want to consider whether collision and comprehensive coverage are cost effective for a 10-year-old automobile. Because they seem to drive relatively little, it may make sense to self-insure.

With sufficient liability coverage on the condominium and the auto, the Perriers may qualify for umbrella liability insurance. They should consider this relatively inexpensive coverage to provide additional liability protection relating to the auto and condominium. As a nurse, Marion may want to consider professional liability insurance if it is not provided by her employer. If either Jacques or Marion serves on any type of board of directors of a corporation, public service, or other entity, they should make sure that that entity provides appropriate liability coverage for its directors.

Longevity Insurance.

Marion's mother may be an excellent candidate for longevity insurance. As noted earlier, longevity insurance is typically structured as a life annuity with payments beginning at some future date, such as age 85. Because it is not underwritten, it is generally a poor choice for someone whose life expectancy is shorter than average for his or her age and a good choice for someone whose life expectancy is longer than average, as is the case with Marion's mother. The annuity payments would not start for more than a decade, but they would offer additional income in her later years and help combat inflation, and the cost would be relatively low. Longevity insurance may be viewed as a complement to or even a partial substitute for long-term care insurance. A cost-benefit analysis of the available options should be performed to ensure that the policy selected best matches the goals and financial resources of the household.

12

THE EFFECT OF HUMAN CAPITAL ON ASSET ALLOCATION AND RISK REDUCTION

- ☐ discuss how asset allocation policy may be influenced by the risk characteristics of human capital
- ☐ recommend and justify appropriate strategies for asset allocation and risk reduction when given an investor profile of key inputs

There are two primary ways to consider how the different subcomponents of an individual's total economic wealth should affect portfolio construction. The first is asset allocation, which includes the overall allocation to risky assets. The second is the underlying asset classes, such as stocks and bonds, selected by the individual. For example, an individual who works in a risky profession that has a high correlation with the stock market might first choose a less aggressive portfolio (e.g., with a lower

allocation to stocks than the average person of the same age). Next, the investor would select which individual stocks and bonds (or asset classes) to hold to minimize the overall risk of the portfolio within a total wealth framework.

For many people and for many occupations, human capital is generally considered to be a relatively bond-like asset. For example, dividend growth in the S&P 500 Index exhibited a quarterly volatility of 16.4% from first quarter 1948 to fourth quarter 2013, whereas wage growth volatility was closer to 2.5% over the same period.⁴ Among individuals, occupation can have a large impact on the degree of wage growth volatility. For example, Oyer (2008) found that stock market conditions can have a strong impact on the lifetime earnings of MBA students. Asset performance that is strongly correlated with the lifetime earnings of a worker will provide less hedging benefit because the assets magnify, rather than reduce, variability in consumption.

EXAMPLE 14

Human Capital and Asset Allocation (1)

1. The riskiness of human capital, as well as that of other assets, should affect the allocation of an individual's financial capital. Consider three investors: George, John, and Sam. Each investor owns only two assets—human capital and financial capital—and wants his total wealth (i.e., human capital plus financial capital) to have a 45% stock allocation. If human capital is assumed to be 30% stock-like, what is the optimal allocation for the financial capital of George, John, and Sam?

Person	Human Capital (HC)	Financial Capital (FC)	Total Wealth (TW)
George	\$500,000	\$150,000	\$650,000
John	\$800,000	\$300,000	\$1,100,000
Sam	\$150,000	\$150,000	\$300,000

Solution:

The allocation is as follows:

	(A)	(B)	(C)	(D)
Person	TW × 45% Target Equity	HC × 30% HC Equity	(A) – (B) FC Equity	(C)/FC FC Equity Allocation %
George	\$292,500	\$150,000	\$142,500	95.00%
John	\$495,000	\$240,000	\$255,000	85.00%
Sam	\$135,000	\$45,000	\$90,000	60.00%

Even within an occupation, each individual has different human capital risks. For example, two people may have the exact same job, but because of random market forces, one could lose his or her job and be unable to find suitable reemployment for an extended period of time. Many financial services workers lost their jobs during the global financial crisis of 2008–2009, and many who became unemployed have been forced to change careers given the lack of available openings. This outcome could have a significant impact on their long-term human capital. Alternatively, a person

⁴ Based on data obtained from the US Bureau of Economic Analysis.

may have a health shock that seriously reduces their ability to rent his or her human capital for the same wage rate as before. Health shocks that meet the threshold of disability may be partially hedged through the insurance market, and the impact of job loss can be partially hedged through unemployment insurance. Most human capital volatility (other than premature death), however, is difficult to hedge through the insurance market.

If both spouses are employed, this may reduce the overall riskiness of the household's human capital. Each spouse provides his or her own income, each with its own risks, but unless the human capital of the two spouses is highly correlated (e.g., if they are both employed in the same family business), their combined human capital will benefit from diversification. An individual who receives income from numerous sources—such as salaries from different jobs, as well as dividends, interest, and so on—must consider the characteristics of his or her total compensation.

A lower-earning partner may also have a more risky human capital value if the higher-earning partner is tied to a specific geographic location. In the event of a job loss, this household may suffer a more significant decline in human capital value than a household in which both partners are able to move to a location where they can maximize their wage rate. The human capital of a less mobile household will have a lower present value and greater volatility. If human capital is very employer-specific (if the individual would have trouble earning the same wage from a different employer), then it is also less valuable and more risky. A household with a non-working spouse may be in a less vulnerable position than a single-person household if the non-working spouse can exercise the option to rejoin the workforce if the primary earner suffers an unexpected loss in earnings.

Most of the wealth an investor holds outside the investment portfolio (e.g., human capital and defined benefit pensions) tends to be relatively conservative in nature (i.e., more bond-like than stock-like). After an optimal investment policy is determined that establishes a target mix of risky and risk-free assets based on risk tolerance, the total wealth asset allocation should be adjusted as the value of the assets changes over time. For example, younger investors should likely allocate more of their investment portfolio to stocks because the value of human capital (which is bond-like) is highest early in the life cycle. Conversely, older investors should shift more of their wealth toward bonds because their bond-like human capital is gradually depleted as they approach retirement. This investment strategy is consistent with such life-cycle investments as target-date funds (in which the target date is the expected retirement date of the individual), which gradually increase the allocation to bonds as investors get closer to retirement.

The economic (holistic) balance sheet discussed earlier in the reading considers the current value of marketable and non-marketable assets. However, this type of balance sheet does not consider the stochastic nature of each individual asset or how the value of one asset rises and falls with respect to other assets within the portfolio over time. It also does not consider the relative liquidity of each asset category, which can be particularly important when there are limited financial instruments that can be used, for example, to borrow against the value of a pension. The allocation for the financial portfolio should be coordinated with the risk associated with the non-marketable assets in an investor's portfolio, such as human capital. Using this perspective, the financial portfolio can be considered a completion or hedge portfolio, because it is invested in such a way as to optimize the overall risk characteristics of an individual's total wealth.

Human capital is a unique asset class in a number of ways. First, it can require a continued investment in knowledge and skills to maintain or increase its value. Some professions are more risky than others and will provide either greater income variance or income that is more strongly correlated with systematic risks (i.e., will rise and fall with economic cycles). Some forms of human capital are more vulnerable to disability risks or premature death. A precision welder may earn the same income

as an accountant, but the welder may be more vulnerable to an injury that would sharply reduce the value of that human capital. Human capital is also more illiquid, and there are limited financial instruments available that can be used to effectively borrow against expected human capital when earnings fall below a desired level of consumption. Some occupations will have an earnings path that is likely to keep pace with inflation, whereas other occupations will not. Employees in some occupations may have invested in human capital that allows them to be very productive in a specific role for their employer. This heavy investment in employer-specific human capital may not easily be rented to another employer for the same wage rate. This places the employee at greater risk of future income volatility. Many who enjoyed steady income growth in the past—for example, mortgage brokers in the United States during the 2000s—were not fully aware of their own human capital risk. Similarly, individuals who work in the real estate industry should likely seek to underweight real estate assets in their portfolios to reduce exposure to real estate from a total wealth perspective. This diversification may be a challenge for those who carry optimistic beliefs about the future growth of their own industry, but history shows that few industries are immune to market shocks.

EXAMPLE 15

Human Capital and Asset Allocation (2)

1. Describe how investment strategies can be modified to account for human capital risk.

Solution:

Investment assets may be strongly or weakly correlated with the human capital value of a worker. The overall volatility of one's economic balance sheet can be reduced by selecting assets that correlate weakly (or even negatively) with human capital. Sector investments may be particularly valuable if they are not a complement to the industry that employs the primary earner. Workers with more volatile assets may also prefer more liquid investments.

EXAMPLE 16

Human Capital and Asset Allocation (3)

1. Compare investment planning for a young family with investment planning for a newly retired couple.

Solution:

A younger household will hold most of its wealth in human capital. For most households, human capital is a bond-like asset that returns a relatively stable income over time. This fact increases the optimal allocation to risky assets within the investment portfolio for younger households. For a newly retired couple, the value of human capital declines relative to the value of the investment portfolio. To balance the total risk of the older household's portfolio, investment portfolio risk should be reduced because investment assets are a larger share of the economic balance sheet (ignoring the impact of charitable bequests and other obligations).

Asset Allocation and Risk Reduction

An individual or household manages wealth risk mainly to smooth spending over time. A strategy that combines appropriate investments with insurance products, or other risk management tools, can be used to provide the highest level of spending for the level of risk the individual or household is willing to take. Each household will have its own risks and preferences that determine which strategy makes sense.

Investment risk, property risk, and human capital risk can be either idiosyncratic or systematic. Idiosyncratic risks include the risks of a specific occupation, the risk of living a very long life or experiencing a long-term illness, and the risk of premature death or loss of property. Within a total wealth framework, idiosyncratic human capital risks are reduced through investment portfolio strategies and/or through insurance (or annuity) products. Pooling risk allows a household to efficiently reduce idiosyncratic risk. Systematic risks affect all households. For example, a diversified investment portfolio of risky assets will be exposed to the systematic risk that the overall market will fall in value. Earnings can also be affected by systematic risk through a recession or slow economic growth. A cure for cancer might increase overall longevity, placing all households at greater risk of outliving their assets.

The first step in creating strategies for asset allocation and risk reduction is to identify idiosyncratic risk exposures that can be efficiently reduced through diversification or hedging. A young doctor with two children and a lower-earning spouse bears a number of idiosyncratic risks. First, the couple's investment portfolio should be well diversified and not highly correlated with the doctor's income. Second, the household's largest asset, its human capital, could be diminished or lost through disability or premature death. Life insurance and disability insurance provide a hedge that pools idiosyncratic human capital risk. Medical malpractice insurance provides protection against idiosyncratic liability risk. In general, hedging these risks trades a small drop in expected wealth for an increase in the likelihood of smooth spending.

Young households may have additional liquidity constraints that will affect recommended products. A doctor who expects his or her earnings to rise sharply and who has high current expenses may choose to defer retirement saving for a few years. This individual should not, however, avoid paying premiums on insurance used to hedge the value of human capital. Term life insurance can provide ample coverage at a modest price, and the household may consider a longer elimination period on disability insurance to fit the cost into the current budget. The investment portfolio allocation may be of little consequence because the value of the total wealth portfolio consists almost entirely of human capital. Instead, investments may be selected to increase liquidity that can be used to protect current consumption from any short-term income shock.

A 60-year-old couple nearing retirement with grown, independent children and a large investment portfolio will face a different set of risks. Although their combined income may be higher than that of the young doctor, the value of their human capital is far less if they plan to retire in five years. Therefore, for the older couple, risks to the investment portfolio are far more relevant than human capital risks. Life insurance may have value only as a means of covering estate planning and liquidity needs. Disability insurance can likely be dropped. Health and liability remain significant idiosyncratic risk exposures that can be efficiently reduced through health, long-term care, and liability insurance. Home equity can be protected through property insurance; the risk of idiosyncratic regional real estate price variation should also be considered. The couple's risk of outliving their assets can be protected through the purchase of annuities. Market risk can be particularly important because, compared with a younger household, the older couple's investments are a far larger share of total household wealth. An older household may have fewer liquidity constraints and thus more flexibility to retain risks—for example, through higher insurance deductibles and longer elimination periods on long-term care insurance. This older household may also

choose to retain the majority of property risk. A rational increase in risk retention can be combined with a reduction in portfolio risk to achieve an efficient balance of total wealth risk. The household will receive a higher expected return (equal to the insurance load) from accepting greater insurable risk while simultaneously reducing the expected return on the investment portfolio by increasing the share of safe assets.

Human capital risks are correlated with market returns and can be at least partially hedged through holistic portfolio construction. Consider a 50-year-old couple, Jennifer and Wade. Jennifer earns US\$75,000 a year working as a tenured college professor, and Wade earns US\$100,000 selling drilling parts to the oil industry. Jennifer participates in the state public employee pension system, which is currently in good enough financial condition to cover promised benefits for the next 40 years. Wade has US\$300,000 saved in a 401(k), half in employer stock. Both are eligible to receive Social Security benefits.

Wade works in a cyclical industry and will see high variability in income but a modest covariance of earnings with the performance of the overall stock market. Although Wade's income today is higher than Jennifer's income, the difference in the present value of their human capital will not be that great. Wade's income is more volatile because it rests largely on the strength of the domestic oil industry. Jennifer may apply a discount rate equivalent to that of current state government bonds because her salary is unlikely to change and her salary risk is comparable to the state's ability to pay general obligation debts. The discount rate on Wade's income may be placed above the historical equity premium because his human capital risk is not easily diversified and is more volatile than the market in general.

To preserve consumption if Wade is laid off for a period of time, the couple should consider holding a significant amount of assets in marketable securities that can be easily liquidated to fund short-term spending needs. These types of assets include money market accounts and short-term government bonds, which can be easily traded in the secondary market. Investment assets held in taxable accounts can be accessed for emergency spending in the event of long-term unemployment. The combination of Jennifer's stable salary and Wade's volatile income allows the couple to take on greater risk than if they lived on Wade's income alone. If Wade and Jennifer have an average level of investment risk tolerance, then they should invest in a mix of stocks and bonds, each within the most tax-efficient account, to create a balanced portfolio appropriate to the couple's risk preferences and the risk of the remainder of the total wealth portfolio.

Wade's more risky human capital is highly correlated with the overall economy, although as previously mentioned, his wages are likely not strongly correlated with the equity market in general. A more efficient investment portfolio will provide a partial hedge against Wade's idiosyncratic human capital risk. The obvious first step would be to sell Wade's investments in employer stock. This may be complicated by restrictions on selling the stock or even Wade's subjective opinion of the potential value of his employer's stock. If Wade is restricted from selling his company stock, an adviser might suggest put options to hedge this risk. To balance the risk to Wade's human capital, the retirement portfolio should overweight sector funds that are either uncorrelated or negatively correlated with the oil industry. Sectors, such as transportation, that perform well when oil prices fall will provide some buffer against fluctuations in the value of Wade's human capital.

The value of Jennifer's pension will likely be a significant share of the couple's total wealth. Discounted at a modest bond rate of comparable duration to the expected pension payments, the value of her pension likely exceeds the US\$300,000 that John has saved in his 401(k). If her pension were less secure, which can be estimated based on the percentage of future obligations that can be funded from current assets, then the pension would be discounted at a higher rate. Jennifer's pension can be viewed as a forced savings plan with a promised stable future payout that is comparable to a bond.

If the discounted value is estimated to be US\$500,000, then this portion of the total portfolio will be characterized as a bond-like asset. Couples with significant pension assets can accept greater risk in the rest of their investment portfolio because poor risky asset performance has less of an impact on future spending when the bulk of consumption will be funded from stable pension income. Wade can take greater risk in his 401(k) because the couple will be able to rely on Jennifer's pension. Note that although it is often difficult to consider, the possibility of divorce should also affect recommendations if pension assets are not allocated to the other spouse. For example, couples may want to hold greater liquidity, and spouses with defined contribution savings or more volatile human capital may choose to take less investment risk.

EXAMPLE 17

Asset Allocation and Risk Reduction

1. Consider two 35-year-old couples, each of which earns a combined US\$150,000 per year. One couple consists of an individual who is employed as a petroleum engineer and a non-working spouse. The other couple consists of two high school teachers. Compare asset allocation and risk reduction strategies for each couple.

Solution:

The human capital value of the couple consisting of the petroleum engineer and the non-working spouse is likely lower than the combined human capital value of the high school teachers, although the combined lifetime cumulative wages of the teachers is likely lower than those of the engineer and the spouse. Earnings for the engineer are highly correlated with oil prices, and either rising or falling prices will affect the household's available income in the future. The impact of a disability on employability may be more severe for the engineer than for a teacher. The engineer should thus likely consider a less risky portfolio and should overweight securities that have a low correlation with oil prices. Conversely, the teachers should select a riskier portfolio as a result of their higher human capital and low correlation with individual market sectors.

SUMMARY

The risk management process for individuals is complex given the variety of potential risks that may be experienced over the life cycle and the differences that exist across households. In this reading, key concepts related to risk management and individuals include the following:

- The two primary asset types for most individuals can be described broadly as human capital and financial capital. Human capital is the net present value of the individual's future expected labor income, whereas financial capital consists of assets currently owned by the individual and can include such items as a bank account, individual securities, pooled funds, a retirement account, and a home.

- Economic net worth is an extension of traditional balance sheet net worth that includes claims to future assets that can be used for consumption, such as human capital, as well as the present value of pension benefits.
- There are typically four key steps in the risk management process for individuals: Specify the objective, identify risks, evaluate risks and select appropriate methods to manage the risks, and monitor outcomes and risk exposures and make appropriate adjustments in methods.
- The financial stages of life for adults can be categorized in the following seven periods: education phase, early career, career development, peak accumulation, pre-retirement, early retirement, and late retirement.
- The primary goal of an economic (holistic) balance sheet is to arrive at an accurate depiction of an individual's overall financial health by accounting for the present value of all available marketable and non-marketable assets, as well as all liabilities. An economic (holistic) balance sheet includes traditional assets and liabilities, as well as human capital and pension value, as assets and includes consumption and bequests as liabilities.
- The total economic wealth of an individual changes throughout his or her lifetime, as do the underlying assets that make up that wealth. The total economic wealth of younger individuals is typically dominated by the value of their human capital. As individuals age, earnings will accumulate, increasing financial capital.
- Earnings risk refers to the risks associated with the earnings potential of an individual—that is, events that could negatively affect someone's human and financial capital.
- Premature death risk relates to the death of an individual, such as a family member, whose future earnings (human capital) were expected to help pay for the financial needs and aspirations of the family.
- Longevity risk is the risk of reaching an age at which one's income and financial assets are insufficient to provide adequate support.
- Property risk relates to the possibility that one's property may be damaged, destroyed, stolen, or lost. There are different types of property insurance, depending on the asset, such as automobile insurance and homeowner's insurance.
- Liability risk refers to the possibility that an individual or other entity may be held legally liable for the financial costs of property damage or physical injury.
- Health risk refers to the risks and implications associated with illness or injury. Health risks manifest themselves in different ways over the life cycle and can have significant implications for human capital.
- The primary purpose of life insurance is to help replace the economic value of an individual to a family or a business in the event of that individual's death. The family's need for life insurance is related to the potential loss associated with the future earnings power of that individual.
- The two main types of life insurance are temporary and permanent. Temporary life insurance, or term life insurance, provides insurance for a certain period of time specified at purchase, whereas permanent insurance, or whole life insurance, is used to provide lifetime coverage, assuming the premiums are paid over the entire period.
- Fixed annuities provide a benefit that is fixed (or known) for life, whereas variable annuities have a benefit that can change over time and that is generally based on the performance of some underlying portfolio or investment.

When selecting between fixed and variable annuities, there are a number of important considerations, such as the volatility of the benefit, flexibility, future market expectations, fees, and inflation concerns.

- Among the factors that would likely increase demand for an annuity are the following: longer-than-average life expectancy, greater preference for lifetime income, less concern for leaving money to heirs, more conservative investing preferences, and lower guaranteed income from other sources (such as pensions).
- Techniques for managing a risk include risk avoidance, risk reduction, risk transfer, and risk retention. The most appropriate choice among these techniques often is related to consideration of the frequency and severity of losses associated with the risk.
- The decision to retain risk or buy insurance is determined by a household's risk tolerance. At the same level of wealth, a more risk-tolerant household will prefer to retain more risk, either through higher insurance deductibles or by simply not buying insurance, than will a less risk-tolerant household. Insurance products that have a higher load will encourage a household to retain more risk.
- An individual's total economic wealth affects portfolio construction through asset allocation, which includes the overall allocation to risky assets, as well as the underlying asset classes, such as stocks and bonds, selected by the individual.
- Investment risk, property risk, and human capital risk can be either idiosyncratic or systematic. Examples of idiosyncratic risks include the risks of a specific occupation, the risk of living a very long life or experiencing a long-term illness, and the risk of premature death or loss of property. Systematic risks affect all households.

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PRACTICE PROBLEMS

The following information relates to questions 1-8

Richard Lansky is an insurance and wealth adviser for individuals. Lansky's first meeting of the day is with Gregory Zavis, age 27, a new client who works as a journalist. Gregory's only asset is \$5,000 in savings; he has \$67,000 in liabilities. During the conversation, Lansky describes the concepts of financial capital and human capital, as well as the components of economic and traditional balance sheets. Gregory asks Lansky:

On which balance sheet are my future earnings reflected?

Gregory does not have medical insurance. He asks Lansky for advice regarding a policy that potentially would allow him to avoid paying for office visits related to minor medical problems.

In the afternoon, Lansky meets with Gregory's parents, Molly and Kirk, ages 53 and 60. Molly is a tenured university professor and provides consulting services to local businesses. Kirk is a senior manager for an investment bank. Lansky determines that Molly's income is more stable than Kirk's.

Kirk and Molly discuss estate planning, and Lansky recommends a whole life insurance policy on Kirk's life, with Molly responsible for paying the premiums. In the event of Kirk's death, Gregory would be entitled to the proceeds from the policy. Lansky explains that one feature of the policy provides for a portion of the benefits to be paid even if a premium payment is late or missed.

Molly tells Lansky that she has recently been reading about annuities and would like to clarify her understanding. Molly makes the following statements.

- | | |
|-------------|--|
| Statement 1 | Both deferred and immediate annuities provide the same flexibility concerning access to invested funds. |
| Statement 2 | The income yield for a given amount invested in a life-only immediate annuity is higher for an older person than for a younger person. |

At the end of the consultation, Molly asks Lansky for advice regarding her retired aunt, Rose Gabriel, age 69. Molly believes that Gabriel's life annuity and pension benefits will provide enough income to meet her customary lifestyle needs. Gabriel lives in her mortgage-free home; her medical insurance plan covers basic health care expenses. Women in Gabriel's family generally have long life spans but often experience chronic health problems requiring extended nursing at home. Therefore, Molly is concerned that medical expenses might exceed Gabriel's net worth during her final years.

1. Gregory's human capital is:
 - A. lower than his financial capital.
 - B. equal to his financial capital.
 - C. higher than his financial capital.

2. The *most* appropriate response to Gregory's balance sheet question is:
 - A. the economic balance sheet only.
 - B. the traditional balance sheet only.
 - C. both the economic and the traditional balance sheets.
 3. Given Gregory's policy preference, which type of medical insurance should Lansky recommend?
 - A. Indemnity plan
 - B. Preferred provider plan
 - C. Health maintenance organization plan
 4. In estimating Molly's human capital value, Lansky should apply an income volatility adjustment that is:
 - A. less than Kirk's.
 - B. the same as Kirk's.
 - C. greater than Kirk's.
 5. Regarding the whole life insurance policy recommended by Lansky, Kirk would be the:
 - A. owner.
 - B. insured.
 - C. beneficiary.
 6. The whole life insurance policy feature described by Lansky is a:
 - A. non-forfeiture clause.
 - B. waiver-of-premium rider.
 - C. guaranteed insurability rider.
 7. Which of Molly's statements about annuities is/are correct?
 - A. Statement 1 only
 - B. Statement 2 only
 - C. Both Statement 1 and Statement 2
 8. The type of insurance that will *best* address Molly's concern about Gabriel is:
 - A. disability insurance.
 - B. longevity insurance.
 - C. long-term care insurance.
-

The following information relates to questions 9-15

Henri Blanc is a financial adviser serving high-net-worth individuals in the United States. Alphonse Perrin, age 55, meets with Blanc for advice about coordinating his employee benefits with his investment and retirement planning strategies. Perrin has adopted a life-cycle portfolio strategy and plans to retire in 10 years. Recently, he received a promotion and \$50,000 salary increase to manage a regional distribution center for a national retail firm. Perrin’s spending needs are currently less than his annual income, and he has no debt. His investment assets consist of \$2,000,000 in marketable securities (90% equity/10% fixed income) and a vineyard with winery valued at \$1,500,000.

Blanc leads Perrin through a discussion of the differences between his financial capital and his human capital, as well as between his traditional balance sheet and his economic balance sheet. Perrin is vested in a defined benefit pension plan based on years of service and prior salary levels. Future benefits will vest annually based on his new salary. Perrin makes the following statements regarding his understanding of pension benefits.

- Statement 1

Unvested pension benefits should be classified as human capital.
- Statement 2

Vested pension benefits should not be classified as financial capital until payments begin.

Perrin asks Blanc to compare his traditional and economic balance sheets. Blanc calculates that the sum of the present values of Perrin’s consumption goals and bequests exceeds that of his unvested pension benefits and future earnings. Perrin tells Blanc that he expects a slower rate of growth in the US economy. Perrin expresses the following concerns to Blanc.

- Concern 1

Holding all else equal, I wonder what the effect will be on my human capital if the nominal risk-free rate declines?
- Concern 2

My employer projects a slower rate of sales growth in my region; therefore, I am anxious about losing my job.

Perrin is a widower with three adult children who live independently. Perrin’s oldest son wishes to inherit the vineyard; the two other children do not want to be involved. Perrin would like to accommodate his children’s wishes; however, he wants each child to inherit equal value from his estate. Blanc explains potential uses of life insurance to Perrin and suggests that one of these uses best meets Perrin’s immediate needs.

Perrin expresses a preference for a life insurance policy that provides a range of investment options. Perrin selects a policy and asks Blanc to calculate the net payment cost index (per \$1,000 of face value, per year), using a life expectancy of 20 years and a discount rate of 5%. Exhibit 1 provides information about Perrin’s policy.

Exhibit 1: Perrin’s Life Insurance Policy	
Face value	\$500,000
Annual premium (paid at beginning of the year)	\$12,000

Policy dividends anticipated per year (paid at end of the year)	\$2,000
Cash value projected at the end of 20 years	\$47,000

9. Which of Perrin's statements regarding his pension is/are correct?
- A. Statement 1 only
 - B. Statement 2 only
 - C. Both Statement 1 and Statement 2
10. Blanc's calculations show that Perrin's economic net worth is:
- A. less than his net worth.
 - B. equal to his net worth.
 - C. greater than his net worth.
11. In response to Perrin's Concern #1, human capital will *most likely*:
- A. decrease.
 - B. remain the same.
 - C. increase.
12. Perrin's Concern #2 identifies a risk related to:
- A. human capital only.
 - B. financial capital only.
 - C. both human and financial capital.
13. Which of the following uses of life insurance *best* meets Perrin's immediate needs?
- A. Provides estate liquidity
 - B. Acts as a tax-sheltered savings instrument
 - C. Replaces lost earning power for dependents
14. The type of life insurance *most appropriate* for Perrin is:
- A. term.
 - B. universal.
 - C. whole life.
15. The net payment cost index that Blanc should calculate is *closest* to:
- A. \$17.48.
 - B. \$20.00.
 - C. \$20.19.
-

The following information relates to questions 16-23

Adrian and Olivia Barksdale live in Australia with their 16-year-old twins. Adrian, 47, works in a highly cyclical industry as an engineering manager at a bauxite mine. Olivia, 46, is an accountant. The Barksdales are saving for their retirement and college funding for both children. Adrian's annual salary is A\$190,000; Olivia's annual salary is A\$85,000. The family's living expenses are currently A\$95,000 per year.

Both Adrian and Olivia plan to work 18 more years, and they depend on their combined income and savings to fund their goals. The Barksdales' new financial adviser, Duncan Smith, recommends an appropriate disability insurance policy to cover Adrian, given his large salary. Because he has a highly specialized job, Adrian is willing to pay for the most comprehensive policy available.

Smith is also concerned about the Barksdales' existing life insurance coverage. Currently, the Barksdales have a term life policy insuring Adrian with a death benefit of A\$100,000. Smith assesses the family's insurance needs in the event Adrian were to die this year. To do so, Smith uses the needs analysis method based on the financial data presented in Exhibit 1 and the following assumptions:

- The discount rate is 6.0%, and the tax rate is 30%.
- Salary and living expenses grow at 3.5% annually.
- Salary and living expenses occur at the beginning of each year.
- The following assumptions apply in the event of Adrian's death:
 - Olivia will continue to work until retirement;
 - Family living expenses will decline by \$30,000 per year;
 - Olivia's projected living expense will be \$50,000 per year for 44 years; and
 - The children's projected living expenses will be \$15,000 per year for 6 years.

Exhibit 1: Barksdale Family Financial Needs Worksheet

Cash Needs	AUD (A\$)
Final expenses and taxes payable	20,000
Mortgage retirement	400,000
Education fund	300,000
Emergency fund	30,000
Total cash needs	750,000
Capital Available	
Cash and investments	900,000
Adrian: Life insurance	100,000
Total capital available	1,000,000

Next, Smith discusses the advantages and disadvantages of annuities. The Barksdales are interested in purchasing an annuity that offers the following

characteristics:

- a payout that begins at retirement,
- the ability to invest in a menu of investment options, and
- a payout that continues as long as either Olivia or Adrian is living.

Olivia's mother, Sarah Brown, is also a client of Smith. She is age 75 and retired, and she needs a known income stream to assist her with current and future expenses. Brown's parents both lived longer than average, and she is concerned about outliving her assets. Smith recommends an annuity.

The Barksdales also worry about longevity risk given their family history and healthy lifestyle. Both spouses want an annuity for their later years (beginning in 40 years) that will ensure the greatest supplemental, level income stream relative to the cost. The Barksdales are willing to forgo the right to cash out the policy.

Smith turns to a discussion about the Barksdales' investment portfolio and how total economic wealth (human capital plus financial capital) might affect asset allocation decisions. The Barksdales' human capital is valued at \$2.9 million and estimated to be 35% equity-like. Smith determines that an overall target allocation of 40% equity is appropriate for the Barksdales' total assets on the economic balance sheet.

Smith makes two recommendations regarding the Barksdales' investment portfolio.

- | | |
|-------------------------|--|
| Recommendation 1 | The portfolio should have lower risk than a portfolio for similar investors in the same lifestyle stage. |
| Recommendation 2 | The portfolio should underweight securities having a high correlation with bauxite demand. |

16. Based on Adrian's job and salary, the *most appropriate* disability policy would define disability as the inability to perform duties of:
 - A. any occupation.
 - B. Adrian's regular occupation.
 - C. any occupation for which Adrian is suited by education and experience.
17. Based on the given assumptions and the data in Exhibit 1, the additional amount of life insurance coverage needed is *closest* to:
 - A. A\$0.
 - B. A\$331,267.
 - C. A\$2,078,101.
18. Based on the Barksdales' annuity preferences, which type of annuity should they purchase?
 - A. Deferred fixed
 - B. Deferred variable
 - C. Immediate variable
19. Based on the Barksdales' annuity preferences, which annuity payout method

should they choose?

- A. Joint life annuity
 - B. Life annuity with refund
 - C. Life annuity with period certain
20. Based on Brown's goals and concerns, which type of annuity should Smith recommend for her?
- A. Deferred fixed
 - B. Immediate fixed
 - C. Immediate variable
21. Which type of annuity *best* satisfies the Barksdales' desire for supplemental income in their later years?
- A. Deferred fixed
 - B. Deferred variable
 - C. Advanced life deferred
22. Based on Exhibit 1, and meeting the Barksdales' target equity allocation for total economic wealth, the financial capital equity allocation should be *closest* to:
- A. 35.0%.
 - B. 54.5%.
 - C. 56.1%.
23. Which of Smith's recommendations regarding the Barksdales' investment portfolio is/are correct?
- A. Recommendation 1 only
 - B. Recommendation 2 only
 - C. Both Recommendation 1 and Recommendation 2
-

SOLUTIONS

1. C is correct. Gregory is in the early career stage of life, and human capital represents a large proportion of his total wealth. Gregory is relatively young; therefore, the present value of his expected earnings implies positive human capital. Furthermore, Gregory's savings are rather low, so his financial capital is small. Consequently, his human capital is greater than his financial capital.
2. A is correct. The present value of expected future earnings is reflected on an economic balance sheet but not on a traditional balance sheet.
3. C is correct. A health maintenance organization plan is a type of medical insurance that allows office visits at no, or very little, cost. Gregory would like to avoid paying for office visits related to minor medical problems; hence this alternative is the most appropriate.
4. A is correct. The income volatility adjustment reflects the fact that income from different professions can vary significantly. Molly works in an industry that has low correlation with the capital markets; she also earns income from an additional source. Kirk works in an industry that has high correlation with capital markets, and so he might experience higher income variability than Molly. Consequently, in estimating Molly's human capital, the income volatility adjustment for Molly should be lower than Kirk's.
5. B is correct. The policy would be on Kirk's life; his death would trigger the insurance payment. Therefore, Kirk would be the insured.
6. A is correct. The whole life insurance policy feature described is a non-forfeiture clause, whereby there is the option to receive some portion of the benefits if premium payments are missed (i.e., before the policy lapses).
7. B is correct. Statement 2 is correct because, all else equal, the income yield is higher when expected longevity is shorter; therefore, the income yield will be higher for an older person.
8. C is correct. Molly is concerned about a potential late-life medical condition that may require extended home care for Gabriel. Long-term care insurance is designed to cover a portion of the cost of home care, assisted living facilities, and/or nursing home expense. Gabriel has enough resources to cover her living expenditures, but her medical insurance might be insufficient to cover the costs of extended home care, medicine, or hospital stays. Consequently, long-term care insurance is the most appropriate insurance choice given Gabriel's situation.
9. A is correct. Unvested pension benefits are typically contingent on future work and are thus considered to be part of human capital. Statement #2 is incorrect: vested pension benefits can be considered components of financial capital.

10. A is correct. Economic net worth is calculated as follows:

Economic net worth = Net worth from the traditional balance sheet + (Present value of future earnings + Present value of unvested pension benefits) – (Present value of consumption goals + Present value of bequests)

Perrin's economic net worth is less than his net worth because the sum of the present values of consumption and bequests is greater than the sum of the present values of future earnings and unvested pensions.

11. C is correct. Human capital, HC_0 , is calculated as follows:

$$HC_0 = \sum_{t=1}^N \frac{p(s_t) w_{t-1} (1 + g_t)}{(1 + r_f + y)^t}$$

Holding all else equal as Perrin directs, a reduction in the nominal risk-free rate, r_f , would decrease the total discount rate, thus increasing the present value of human capital.

12. C is correct. The projected slowdown in his employer's sales growth may result in Perrin's unemployment, indicating that he may be subject to earnings risk. Human capital would be reduced by the loss of future earnings and halt accrual of pension benefits at Perrin's present employer. Financial capital could also be affected because assets may need to be sold to make up for any loss of income.
13. A is correct. Life insurance best meets Perrin's immediate need for estate liquidity. A life insurance policy can provide liquidity without the delay involved in the legal process of settling the estate. This liquidity can be particularly valuable if the estate contains illiquid assets or assets that are difficult to separate and distribute equitably among heirs. Currently, it would be difficult to separate and equitably distribute Perrin's financial assets to his three children such that the oldest son inherits the vineyard and winery while keeping the other two children uninvolved because the business is worth more than one-third of Perrin's investment assets. The problem of separating and equitably distributing the estate exists presently regardless of the value of Perrin's personal property.
14. B is correct. Perrin's estate distribution plan indicates a need for estate liquidity funded by permanent insurance that can remain in force until his death. Perrin prefers a policy that offers a range of investment options. Universal life is thus most appropriate because it is a form of permanent insurance that can remain in force until Perrin's death and typically has more options for investing the cash value than do whole life policies.
15. C is correct. The net payment cost index assumes that the insured will die at the end of a specified period—in this case, the given life expectancy of 20 years. Calculating the net payment cost index includes the following steps.

Future value of premiums (annuity due, 5%, 20 years) \$416,631.02

Financial calculator operations:

$N = 20, I = 5, PV = 0, PMT = -12,000, \text{mode} = \text{begin}; FV \rightarrow 416,631.02$

Future value of dividends (ordinary annuity, 5%, 20 years) (\$66,131.91)

$N = 20, I = 5, PV = 0, PMT = 2,000, \text{mode} = \text{end}; FV \rightarrow -66,131.91$

20-Year insurance cost \$350,499.11

Annual payments for insurance cost (annuity due, 5%, 20 years) (\$10,095.24)

$N = 20, I = 5, PV = 0, FV = 350,499.11, \text{mode} = \text{begin}; \text{PMT} \rightarrow -10,095.24$

Net payment cost index (\$10,095.24/500) (\$20.19)

16. B is correct. The most comprehensive policy would define disability as the inability to perform Adrian's regular occupation. For professionals with specialized skills, policies that use regular occupation are generally preferred even though they are more expensive. Mr. Barksdale works in a specialized, high-paying occupation, and the family depends on his income.

17. B is correct. The additional amount of life insurance coverage needed is calculated as the difference between the family's total financial needs and total capital available.

Total financial needs are calculated as follows.

Cash Needs	AUD (A\$)
Final expenses and taxes	20,000
Mortgage retirement	400,000
Education fund	300,000
Emergency fund	30,000
Total cash needs	750,000
Capital Needs	Present Value
Olivia's living expenses, 44 years	1,377,175
Children's living expenses, 6 years	84,848
(Olivia's income, 18 years)	-880,756
Total capital needs	581,267
Total financial needs	1,331,267

Capital needs are determined as the present value of an annuity due: growth rate = 3.5%, discount rate = 6.0%. Growth of payments is incorporated by adjusting the discount rate to account for the growth rate of earnings. As long as the discount rate is larger than the growth rate, the adjusted rate i can be calculated as follows: $[(1 + \text{Discount rate}) / (1 + \text{Growth rate})] - 1$, or $i = (1.06 / 1.035) - 1 = 2.42\%$.

The present value of Olivia's living expenses is calculated as follows:

$\text{PMT} = -\$50,000; i = 2.42\%, n = 44$. Set for payments at beginning of year. $\text{PV} = \$1,377,175$.

The present value of the children's living expenses is calculated as follows:

$\text{PMT} = -\$15,000; i = 2.42\%, n = 6$. Set for payments at beginning of year. $\text{PV} = \$84,848$.

The present value of Olivia's income is calculated as follows:

$PMT = -\$85,000 \times (1 - \text{Tax rate})$; $PMT = \$85,000 \times 0.70 = 59,500$; $i = 2.42\%$, $n = 18$. Set for payments at beginning of year. $PV = -\$880,756$.

Total capital needs are calculated as follows:

$\$1,377,175 + \$84,848 - \$880,756$

$= \$581,267$. Adding this amount to total cash needs of \$750,000 results in total financial needs of \$1,331,267.

The total capital available is calculated as follows.

Capital Available	AUD (A\$)
Cash and investments	900,000
Current life insurance	100,000
Total capital available	1,000,000

The additional life insurance need is calculated as follows.

Total financial needs	1,331,267
Total capital available	1,000,000
Life insurance shortfall (excess)	331,267

18. B is correct. The Barksdales want an annuity with a deferred payout (beginning at retirement) and an ability to invest in a diversified mix of securities. Most deferred variable annuities offer a diversified menu of potential investment options, whereas a fixed annuity locks the annuitant into a portfolio of bond-like assets at whatever rate of return exists at the time of purchase.
19. A is correct. A joint life annuity best addresses the Barksdales' goal of receiving a payout as long as either of them is alive. Under a joint life annuity, two or more individuals, such as a husband and a wife, receive payments until all beneficiaries die.
20. B is correct. With immediate fixed annuities, Brown will trade a sum of money today for a promised income benefit for as long as she is alive. Brown is already age 75 and is concerned about longevity risk; she wants a known income stream currently and in the future. Therefore, an immediate fixed annuity is the most appropriate choice.
21. C is correct. In contrast to an immediate payout annuity, an advanced life deferred annuity's (ALDA's) payments begin later in life—for example, when the individual turns 80 or 85. An ALDA would provide the greatest supplemental level income relative to the cost because the payments are made far in the future, life expectancy is shorter when the payments begin, and some policyholders will die without receiving payments.
22. C is correct. The equity allocation of the Barksdale's financial capital is calculated as follows:

$$\begin{aligned} \text{Total economic wealth} &= \text{Human capital} + \text{Financial capital} = \$2,900,000 + \$900,000 \\ &= \$3,800,000. \end{aligned}$$

$$\begin{aligned} \text{Target equity allocation of total economic wealth} &= \$3,800,000 \times 40\% \\ &= \$1,520,000 \end{aligned}$$

$$\text{Human capital equity allocation} = \$2,900,000 \times 35\% = \$1,015,000$$

$$\text{Financial capital equity allocation} = \$1,520,000 - \$1,015,000 = \$505,000$$

$$\begin{aligned}\% \text{ Financial capital equity allocation} &= \text{Financial equity allocation} / \text{Total financial capital} \\ &= \$505,000 / \$900,000 \\ &= 0.5611, \text{ or } 56.1\%\end{aligned}$$

23. C is correct. People with higher risk and potential volatility in income (human capital) should take on lower risk in their investment portfolios. Adrian's income is more than two-thirds of the household total and is somewhat volatile because of cyclical demand for his employer's product. Additionally, because income is tied to a particular industry or sector, the Barksdales should underweight securities having a high correlation with bauxite demand.